Université de Lille

15/09/2026

L2 EEA Systèmes Electriques

Régime sinusoïdal

Y Le Menach

FST / EEA / Energie Electrique
Y Le Menach

Définitions de la puissance électrique

- La puissance électrique que l'on note souvent P et qui a pour unité le watt (symbole W) est le produit de la tension électrique aux bornes de laquelle est branchée l'appareil (en volts) et de l'intensité du courant électrique qui le traverse (en ampères) pour des appareils purement résistifs. (Wikipédia)

  En régime continu

$$P = UI$$

- Si la tension et le courant varient, la puissance instantanée consommée par un dipôle est égale au produit des valeurs instantanées du courant qui le traverse et de la tension à ses bornes.

$$p(t) = u(t) \cdot i(t)$$

Définitions

Notion de grandeurs efficaces

Effet Joule :

La loi de Joule décrit le phénomène appelé l'effet Joule :
l'énergie calorifique E (en joule) dégagée par un conducteur électrique de résistance R (en ohm) traversé par un courant d'intensité I (A) pendant un temps t (S) est donnée par la relation suivante :

$$W = R\int_{t_1}^{t_2} i^2(t)\ \mathrm{d}t \qquad P = \frac{W}{t_2 - t_1} = \frac{R}{t_2 - t_1}\int_{t_1}^{t_2} i^2(t)\ \mathrm{d}t$$

Si I est constant

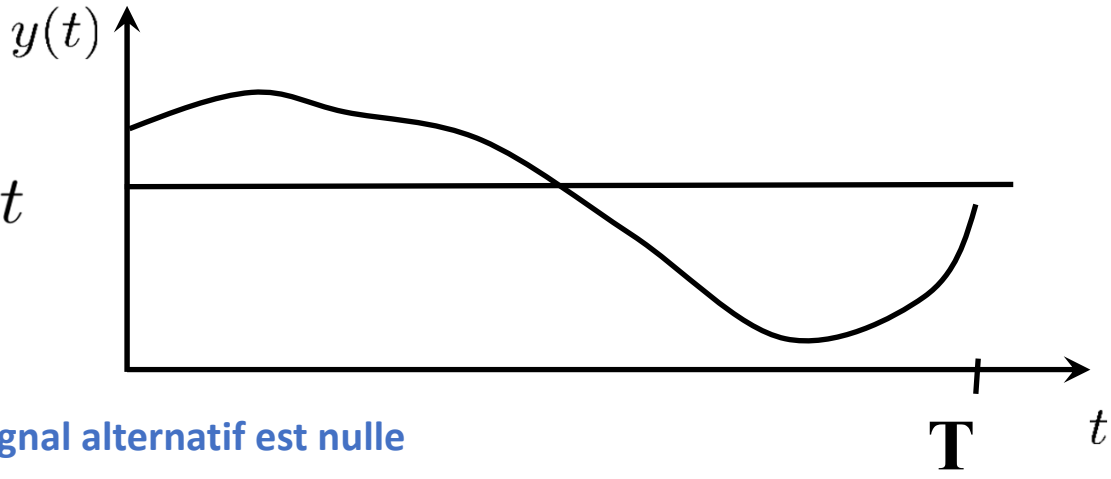
$$P = \frac{R}{t_2 - t_1}\Big[I^2\ t\Big]_{t_1}^{t_2} = \frac{RI^2(t_2 - t_1)}{t_2 - t_1} = RI^2$$

La valeur efficace d'un signal est la valeur du signal continu qui produirait le même effet Joule dans le même conducteur.

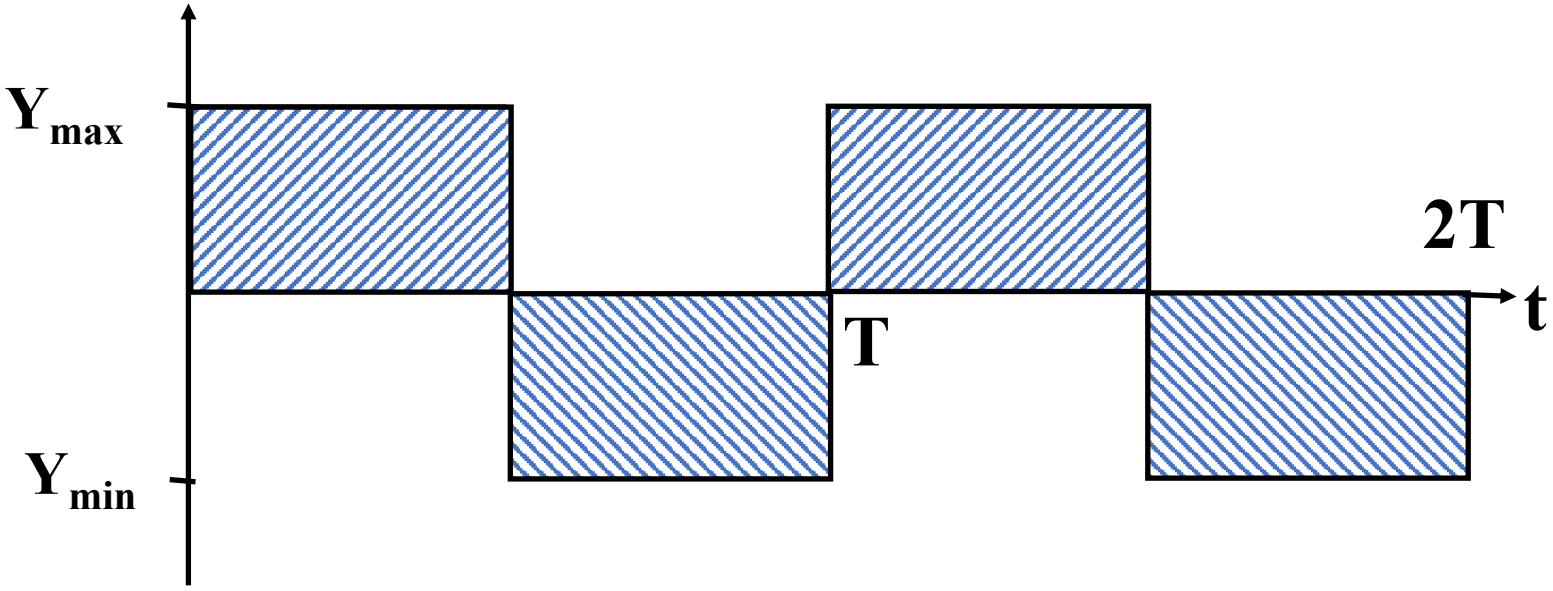
Définitions

- Notion de grandeurs moyennes

$$Y_{moy} = \frac{1}{T}\int_T y(t)\ \mathrm{d}t$$

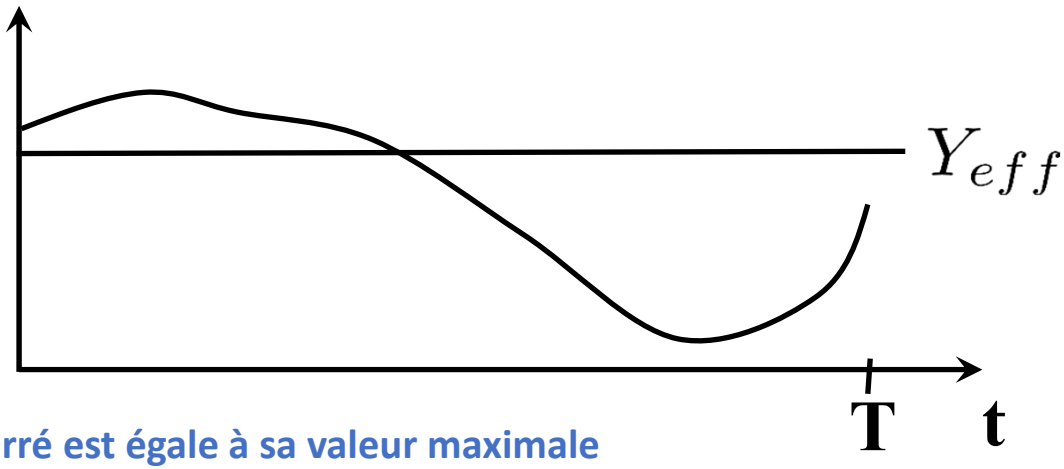


La valeur moyenne d'un signal alternatif est nulle

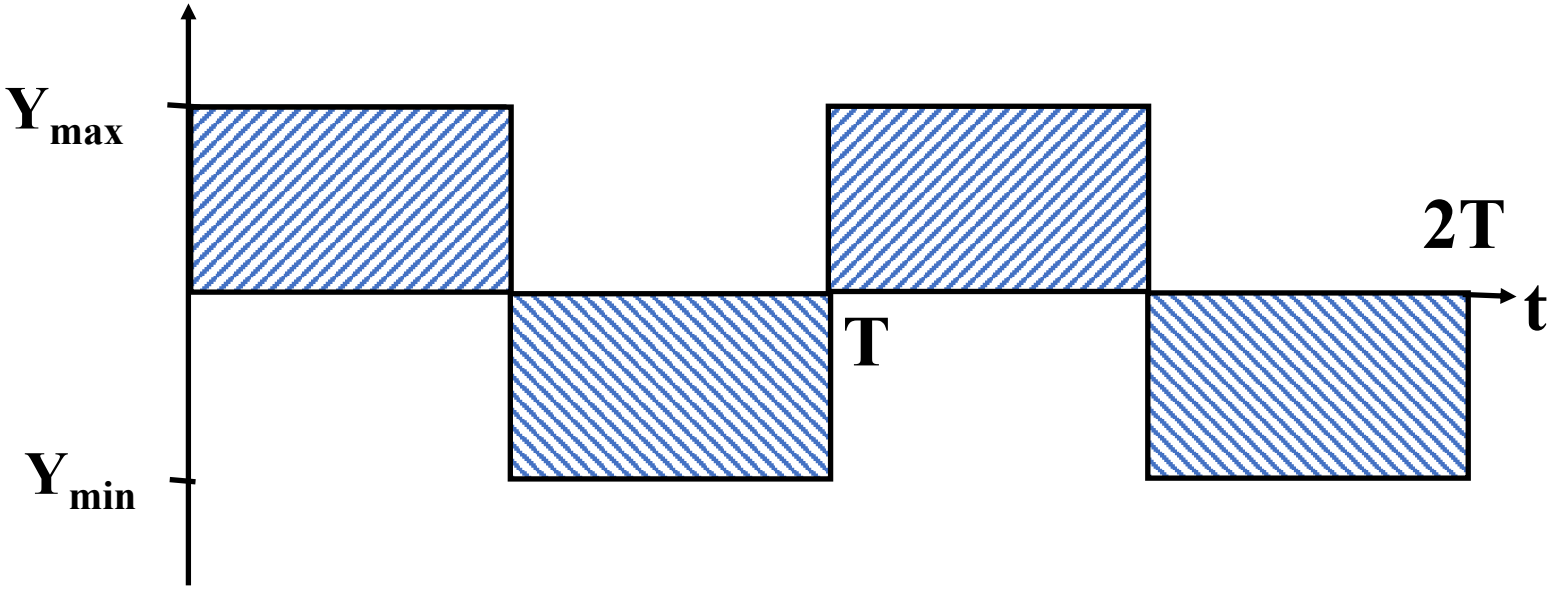


- Notion de grandeur efficace

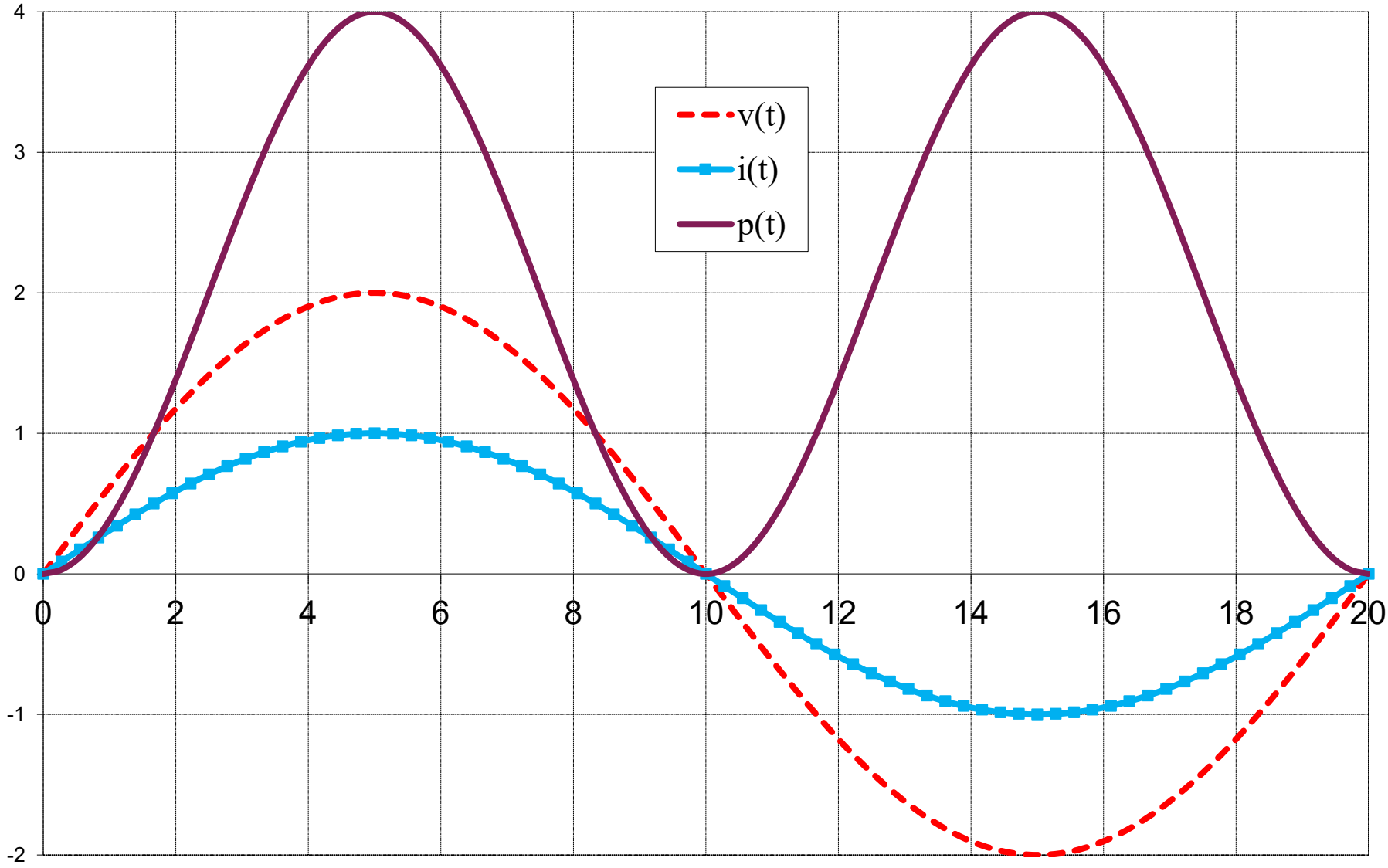
$$Y_{eff}^2 = \frac{1}{T}\int_T y(t)^2 \, \mathrm{d}t$$



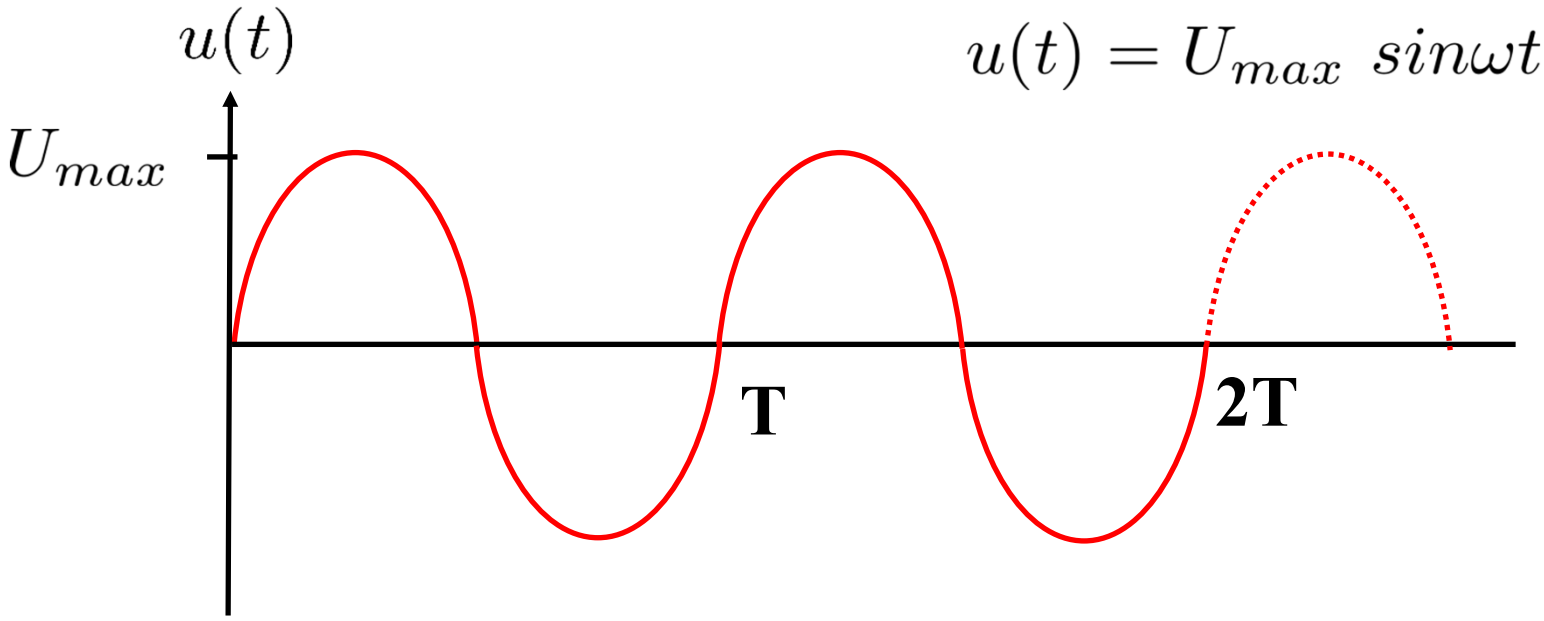
La valeur efficace d'un signal carré est égale à sa valeur maximale



Régime sinusoïdal



- Valeur moyenne d'une grandeur sinusoïdale



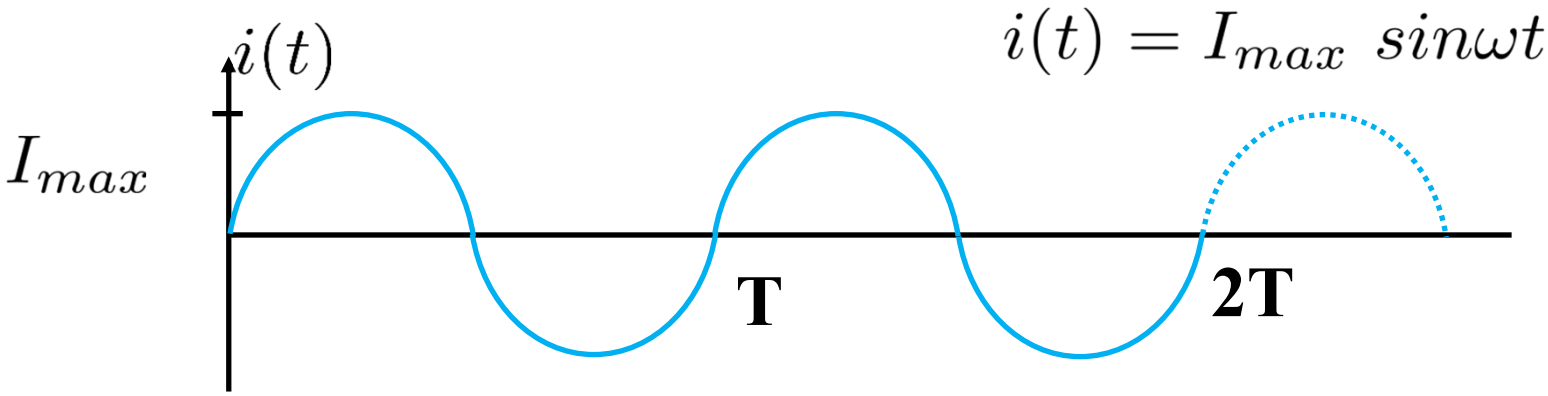
$$Umoy = \frac{1}{T}\int_T U_{max}\ sin\omega t\ \mathrm{d}t \text{ avec la pulsation } \omega = 2\pi f = \frac{2\pi}{T}$$

$$U_{moy} = \frac{U_{max}}{T\omega}\Big[-cos\omega t\Big]_0^T = \frac{U_{max}}{2\pi}\big(-cos(2\pi) + cos(0)\big)$$

$$U_{moy} = 0$$

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- Valeur efficace d'une grandeur sinusoïdale



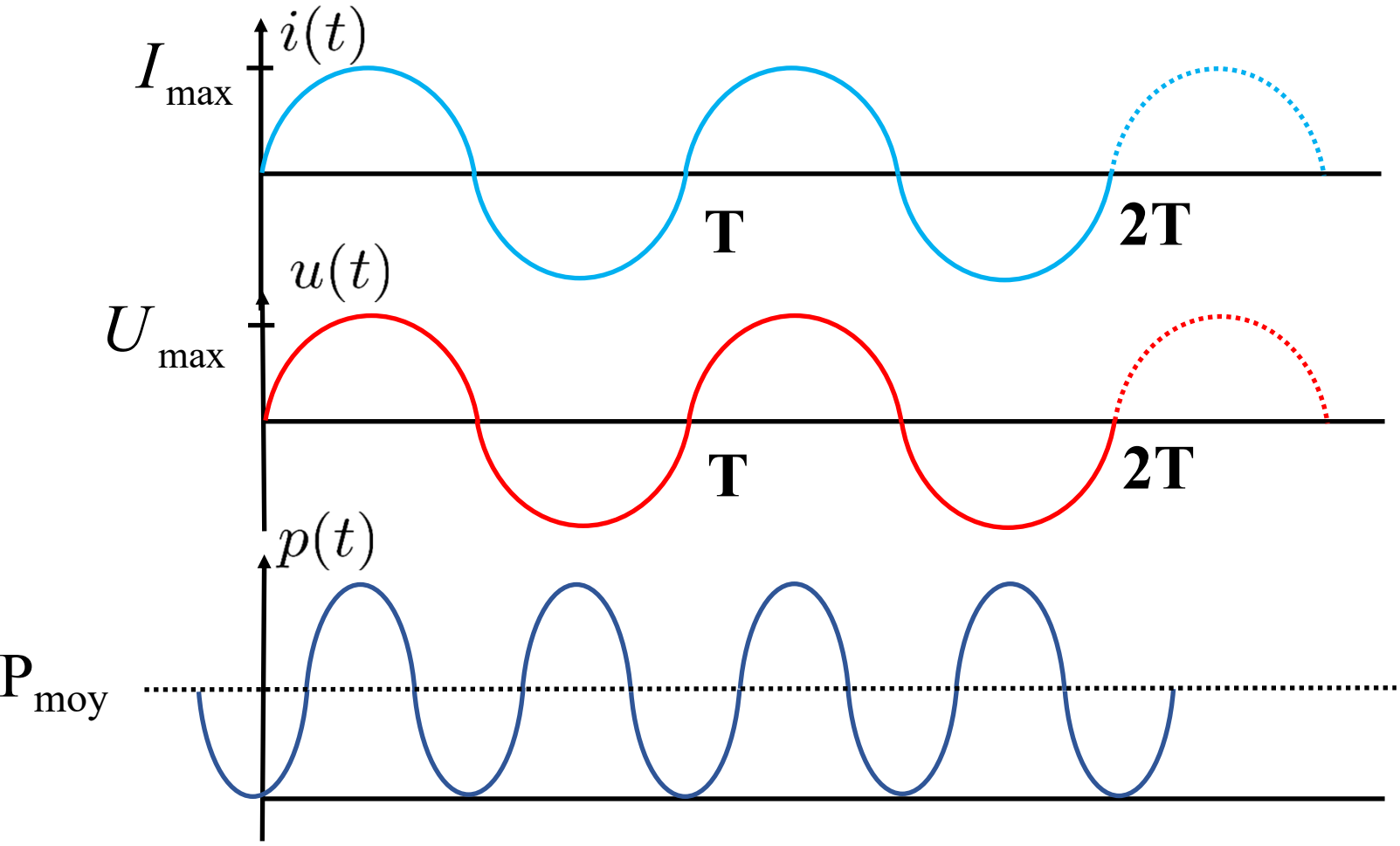
$$P = RI_{eff}^2 \qquad I_{eff}^2 = \frac{1}{T}\int_T i(t)^2\ \mathrm{d}t$$

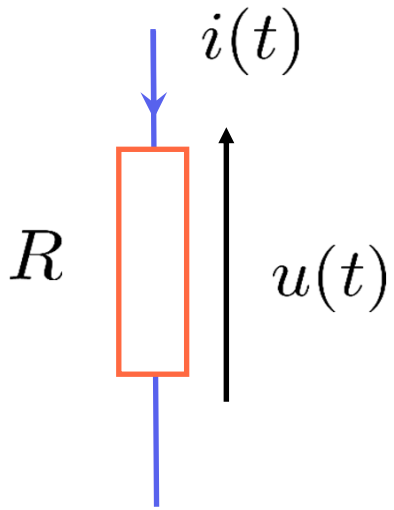
$$I_{eff}^2 = \frac{1}{T}\int_T I_{max}^2\ sin^2\omega t\ \mathrm{d}t = \frac{I_{max}^2}{T}\int_T \frac{1 - cos2\omega t}{2}\ \mathrm{d}t$$

$$I_{eff}^2 = \frac{I_{max}^2}{2T}\left[t - \frac{sin2\omega t}{2\omega}\right]_0^T = \frac{I_{max}^2}{2} \Rightarrow I_{eff} = \frac{I_{max}}{\sqrt{2}}$$

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Puissance dissipée dans une résistance en régime sinusoïdale





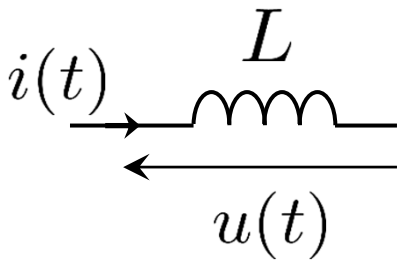
$$u(t) = R\ i(t)$$

$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = \frac{1}{T}\int_T U_{max} sin\omega t \cdot I_{max} sin\omega t\ \mathrm{d}t \quad \Rightarrow \quad P = U_{eff}\cdot I_{eff}$$

$$P = \frac{U_{max}\cdot I_{max}}{T}\int_T sin^2\omega t\ \mathrm{d}t = \frac{U_{max}\cdot I_{max}}{2T}\left[t - \frac{sin2\omega t}{2\omega}\right]_0^T = \frac{U_{max}\cdot I_{max}}{2}$$

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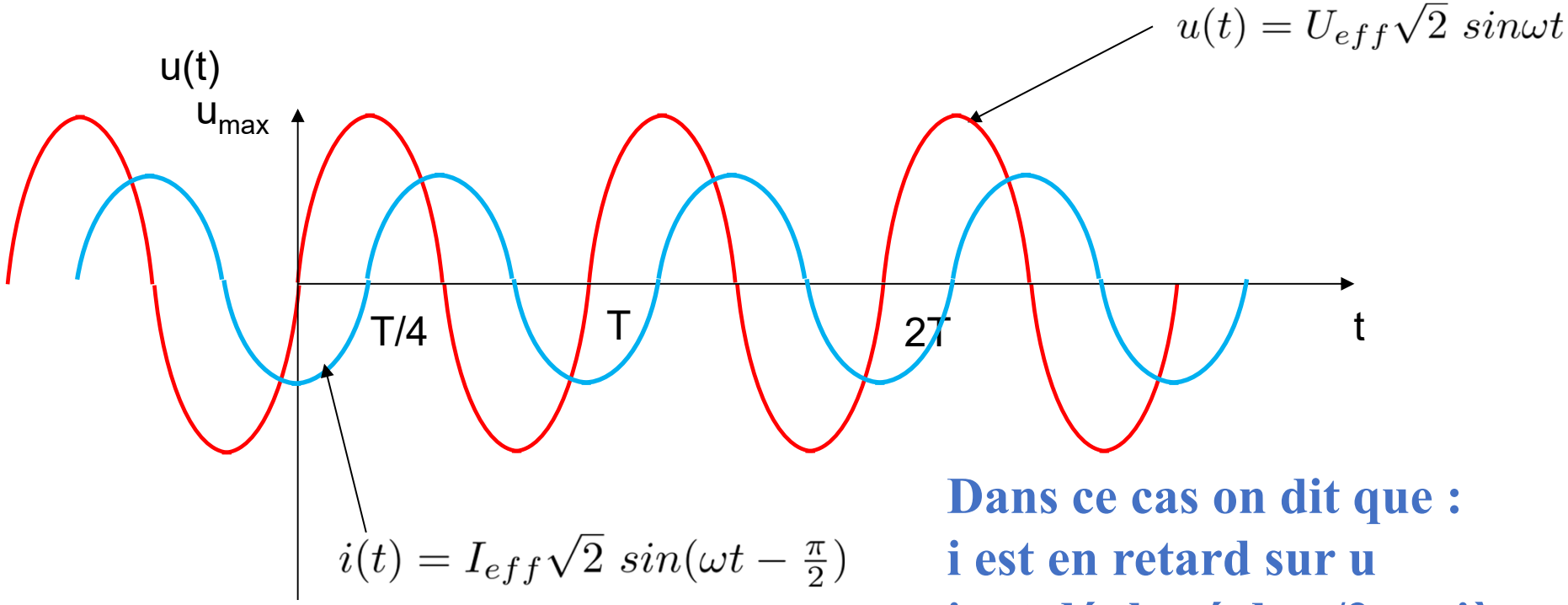
La Bobine



L est la valeur de l'inductance de la bobine
L est exprimée est Henry [H]

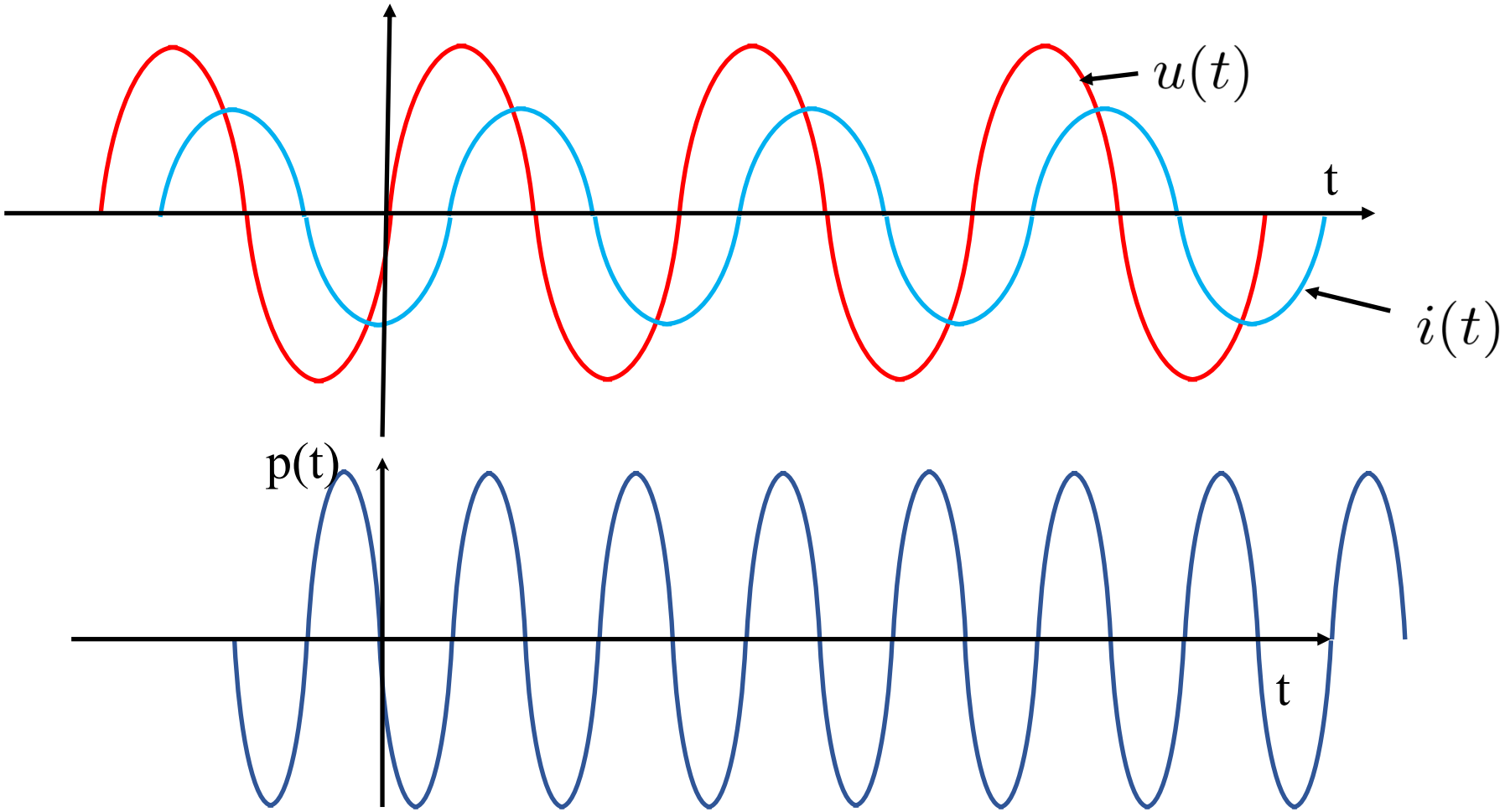
$$u(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t} \Rightarrow i(t) = \frac{1}{L}\int u(t)$$

$$i(t) = \frac{1}{L}\int U_{eff}\sqrt{2}\ sin\omega t = -\frac{U_{eff}\sqrt{2}}{L\omega}\ cos\omega t$$



Dans ce cas on dit que :
i est en retard sur u
i est déphasé de π/2 arrière

Puissance dans une bobine en régime sinusoïdale



$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = -\frac{1}{T}\int_T U_{eff}\sqrt{2}sin\omega t \cdot I_{eff}\sqrt{2}cos\omega t\ \mathrm{d}t$$

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Puissance dans une bobine en régime sinusoïdale

$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = -\frac{1}{T}\int_T U_{eff}\sqrt{2}sin\omega t \cdot I_{eff}\sqrt{2}cos\omega t\ \mathrm{d}t$$

$$P = -\frac{2\cdot U_{eff}I_{eff}}{T}\int_T sin\omega t\cdot cos\omega t\ \mathrm{d}t$$

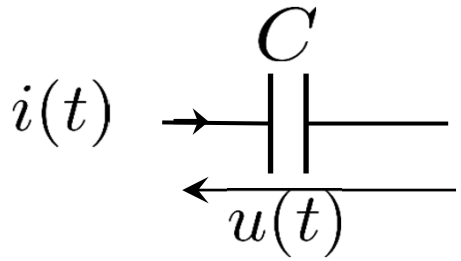
$$\text{comme } 2\ sinx\cdot cox = sin2x$$

$$P = -\frac{U_{eff}I_{eff}}{T}\int_T sin2\omega t\ \mathrm{d}t$$

$$P = -\frac{U_{eff}I_{eff}}{2T}\Big[-cos2\omega t\Big]_0^T = 0$$

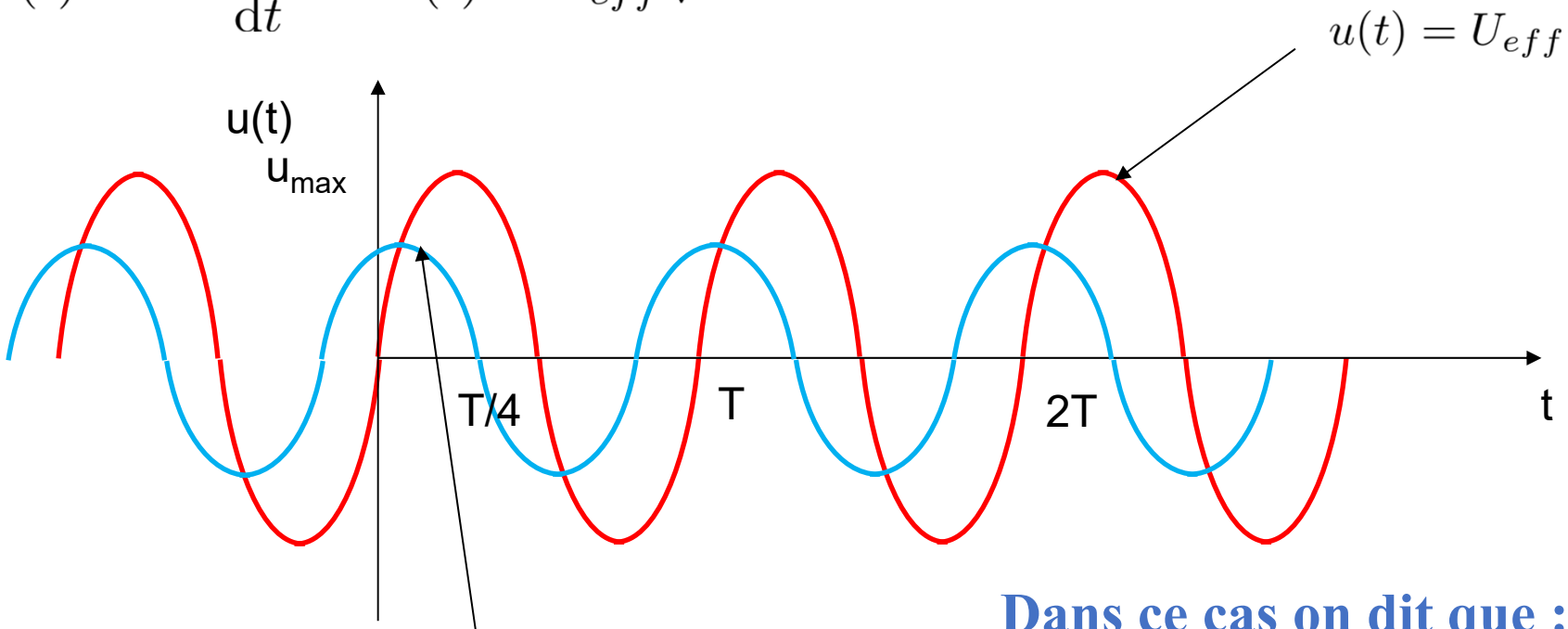
En régime sinusoïdal, la puissance dissipée dans un bobine est nulle

Le Condensateur



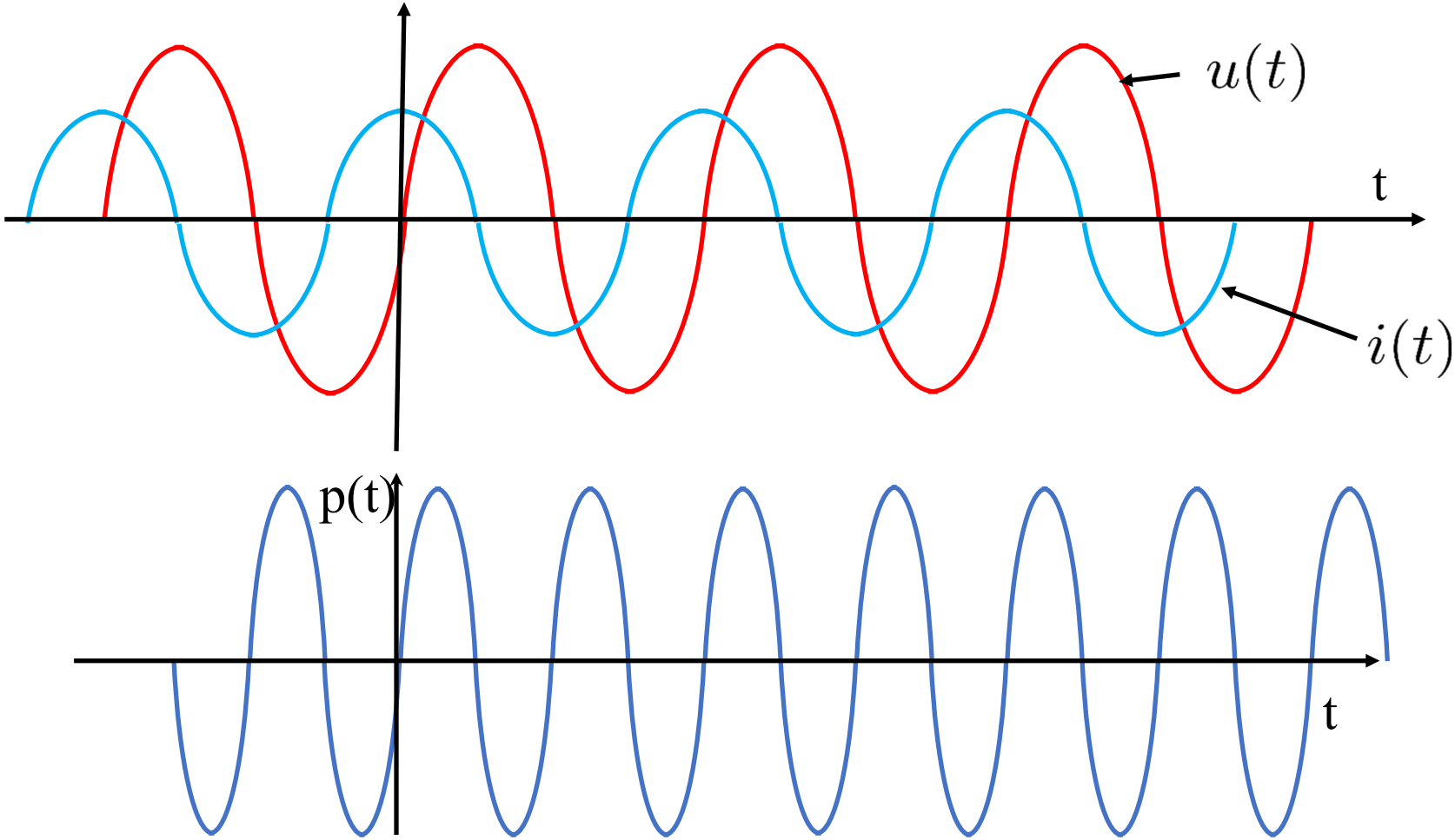
C est la valeur de la capacité du condensateur
C est exprimé est Farad [F]
(usuellement en µF)

$$i(t) = C\frac{\mathrm{d}u(t)}{\mathrm{d}t} \Rightarrow i(t) = U_{eff}\sqrt{2}C\omega\ cos\omega t$$



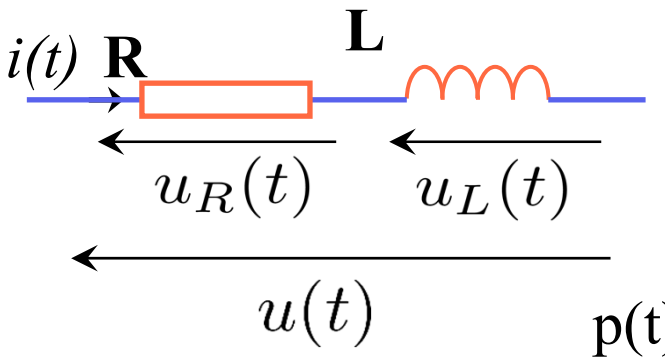
Dans ce cas on dit que :
i est en avance sur u
i est déphasé de π/2 avar

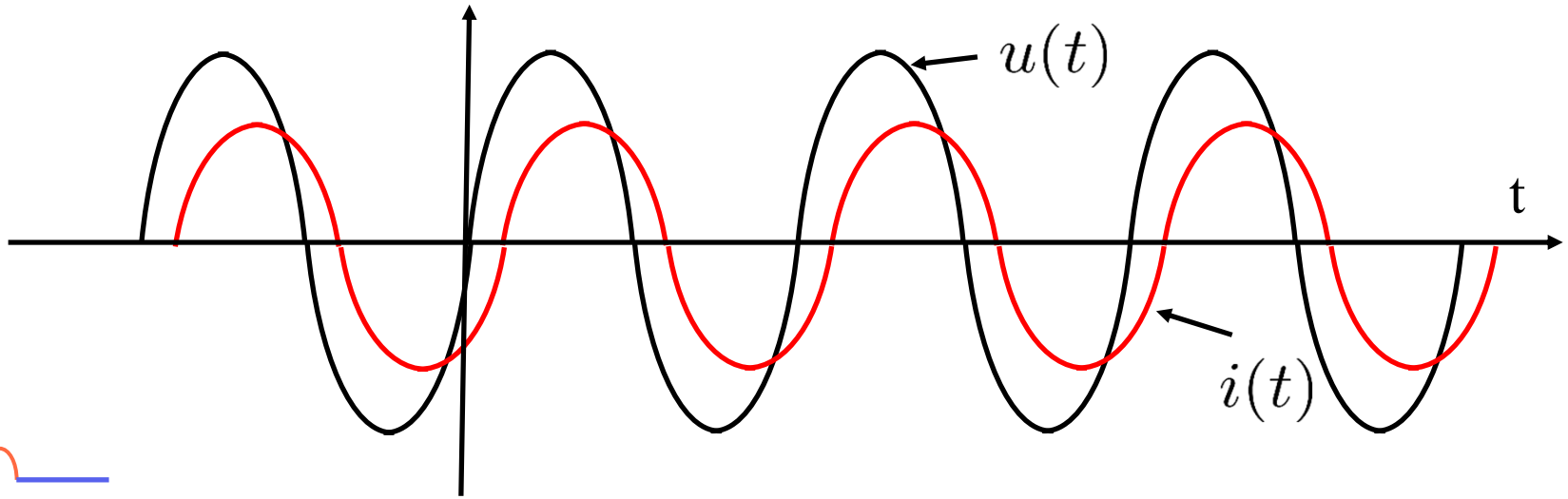
Puissance dans un condensateur en régime sinusoïdale

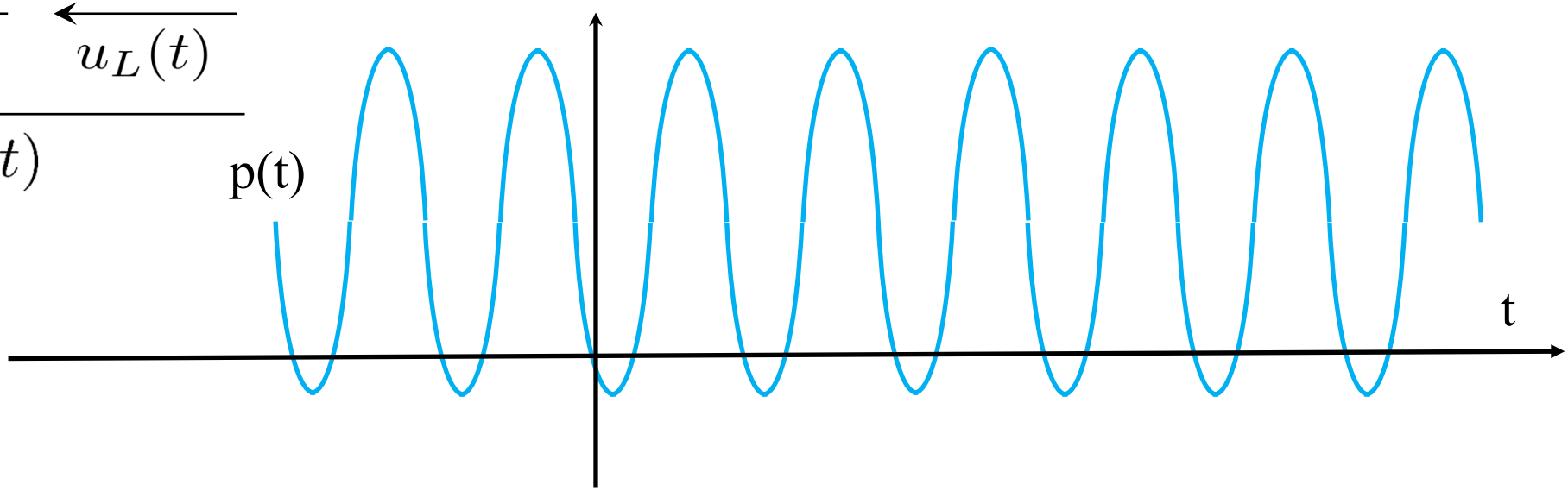


$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = \frac{1}{T}\int_T U_{eff}\sqrt{2}sin\omega t \cdot I_{eff}\sqrt{2}cos\omega t\ \mathrm{d}t = 0$$

Régime sinusoïdale : Circuirt RL







$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = \frac{1}{T}\int_T U_{eff}\sqrt{2}sin\omega t \cdot I_{eff}\sqrt{2}sin(\omega t - \varphi)\ \mathrm{d}t$$

$$P = \frac{1}{T}\int_T u(t)\cdot i(t)\ \mathrm{d}t = \frac{1}{T}\int_T U_{eff}\sqrt{2}sin\omega t \cdot I_{eff}\sqrt{2}sin(\omega t - \varphi)\ \mathrm{d}t$$

$$P = \frac{2\cdot U_{eff}I_{eff}}{T}\int_T sin\omega t \cdot sin(\omega t - \varphi)\ \mathrm{d}t$$

$$\text{comme}\quad sina \cdot sinb = \frac{1}{2}(cos(a-b) + cos(a+b))$$

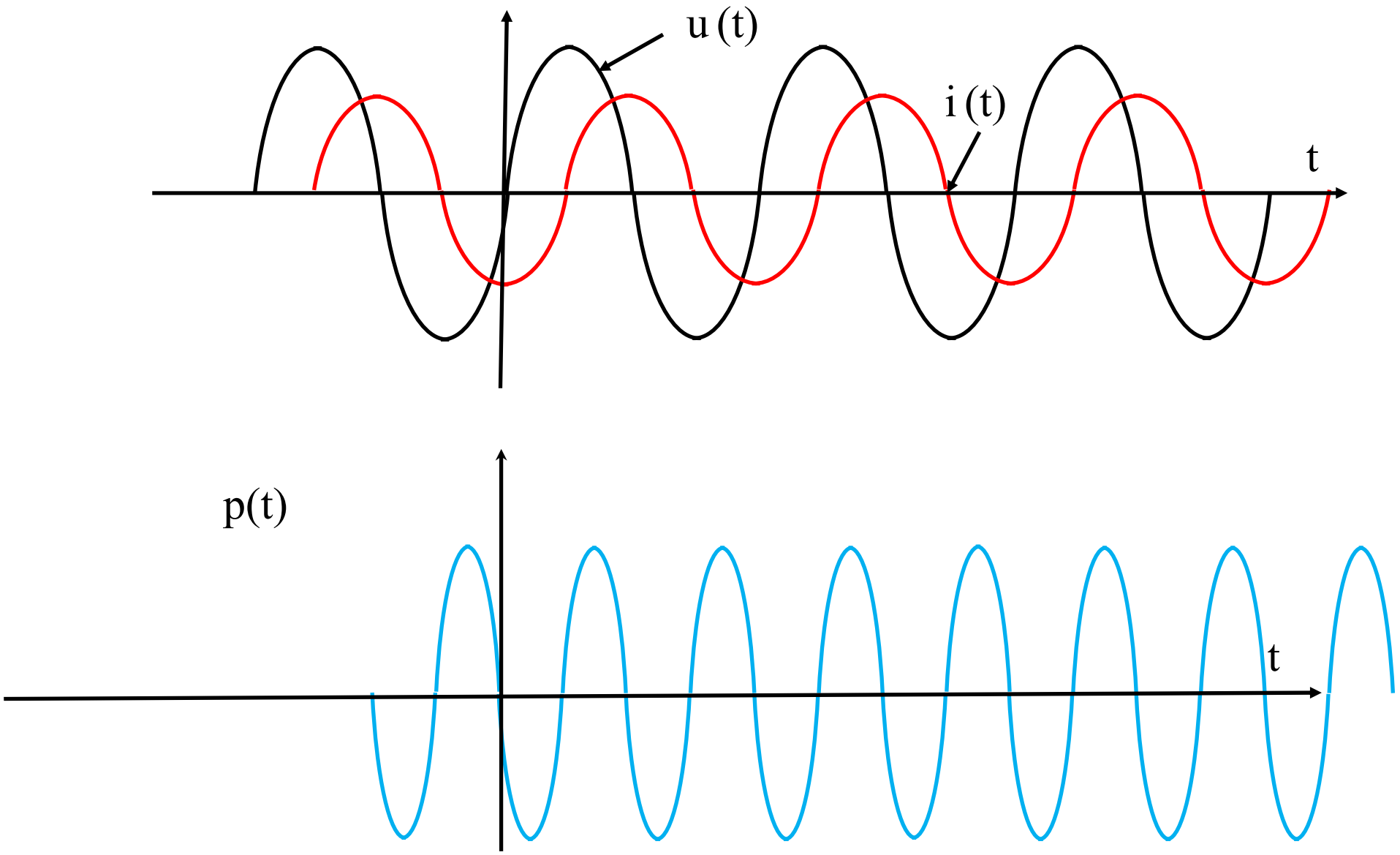
$$\text{alors}\quad sin\omega t \cdot sin(\omega t - \varphi) = \frac{1}{2}(cos\varphi + cos(2\omega t - \varphi))$$

$$P = \frac{U_{eff}I_{eff}}{T}\left(\int_T cos\varphi\ \mathrm{d}t - \int_T cos(2\omega t - \varphi)\ \mathrm{d}t\right)$$

$$P = U_{eff}I_{eff}cos\varphi$$ **P est appelée puissance active**

$cos\varphi$ **est le facteur de puissance du montage**

Puissance en régime sinusoïdal



Puissance apparente en Volt-Ampère[VA]

$$S = U_{eff} I_{eff}$$

Puissance réactive en Volt-Ampère réactif [Var]

$$Q = U_{eff} I_{eff} sin\varphi$$

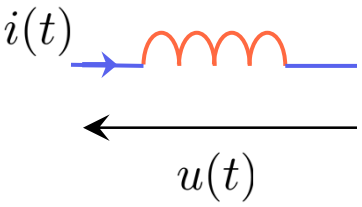
Si le circuit est inductif Q est positif ; le montage consomme de la puissance réactive

Si le circuit est capacitif Q est négatif ; le montage fournit de la puissance réactive

$$\underline{S} = P + jQ = U_{eff} I_{eff} (cos\varphi + jsin\varphi) = \underline{U} \cdot \underline{I}^*$$

$$S = \sqrt{P^2 + Q^2}$$

Puissance réactive dans une bobine et un condensateur



$$u(t) = L\frac{\mathrm{d}i(t)}{\mathrm{d}t}$$

$$\Rightarrow i(t) = \frac{1}{L}\int u(t)\mathrm{d}t$$

si $u(t) = U_{eff}\sqrt{2}\ sin\omega t$

$$\Rightarrow i(t) = \frac{U_{eff}\sqrt{2}}{L\omega}\ sin(\omega t - \tfrac{\pi}{2})$$

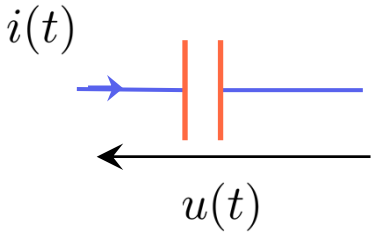
$$\varphi = \frac{\pi}{2} \Rightarrow cos\varphi = 0 \text{ et } sin\varphi = 1$$

$$P = 0 \text{ et } Q = U_{eff} \cdot I_{eff}$$

comme $\underline{U} = jL\omega\underline{I} \Rightarrow \|\underline{U}\| = \|jL\omega\| \cdot \|\underline{I}\|$

$$Q = L\omega \cdot I_{eff}^2$$

$$Q = \frac{U_{eff}^2}{L\omega}$$



$$i(t) = C\frac{\mathrm{d}u(t)}{\mathrm{d}t}$$

si $u(t) = U_{eff}\sqrt{2}\ sin\omega t$

$$\Rightarrow i(t) = C\omega U_{eff}\sqrt{2}\ sin(\omega t + \tfrac{\pi}{2})$$

$$\varphi = -\frac{\pi}{2} \Rightarrow cos\varphi = 0 \text{ et } sin\varphi = -1$$

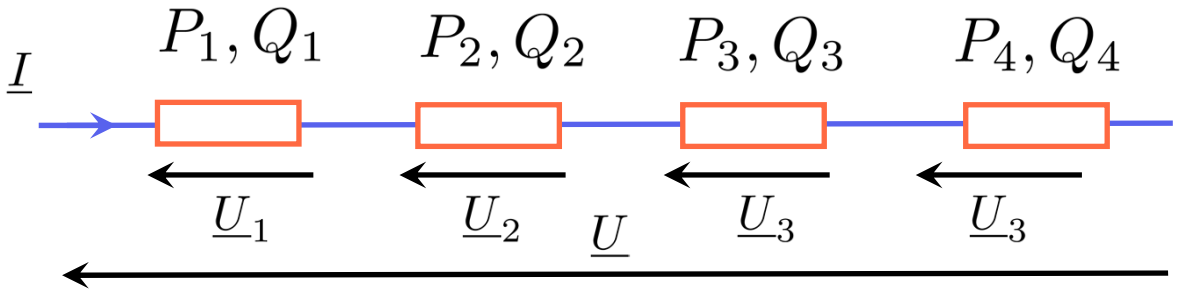
$$P = 0 \text{ et } Q = -U_{eff} \cdot I_{eff}$$

comme $\underline{U} = \frac{\underline{I}}{jC\omega} \Rightarrow \|\underline{U}\| = \frac{\|\underline{I}\|}{\|jC\omega\|}$

$$Q = -C\omega \cdot U_{eff}^2$$

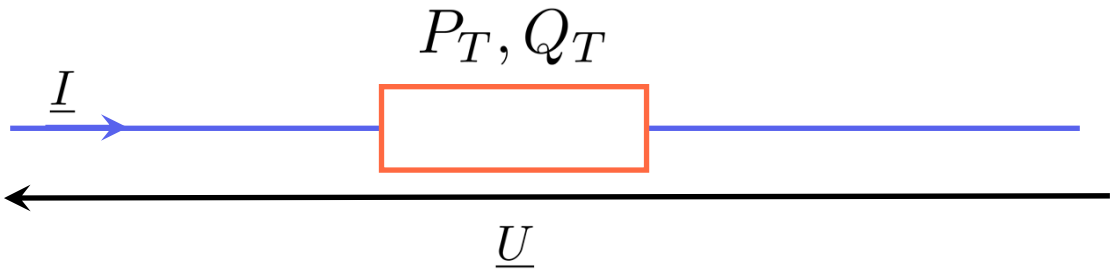
$$Q = -\frac{I_{eff}^2}{C\omega}$$

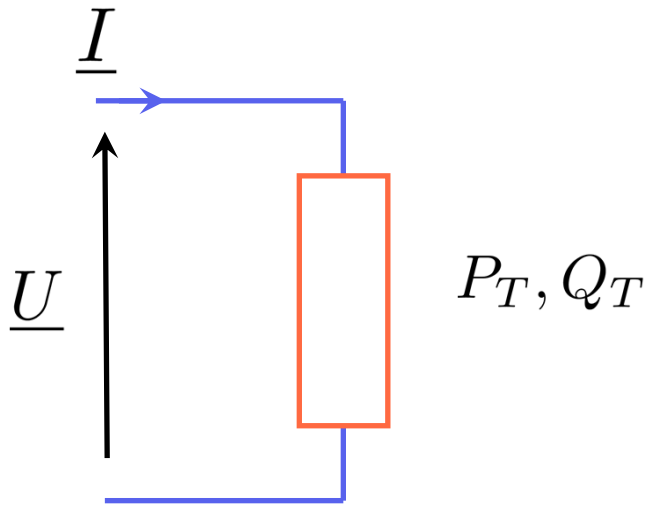
Méthode de Boucherot



$$\underline{U} = \underline{U_1} + \underline{U_2} + \underline{U_3} + \underline{U_4}$$

$$\|\underline{U}\| \neq \|\underline{U_1}\| + \|\underline{U_2}\| + \|\underline{U_3}\| + \|\underline{U_4}\|$$





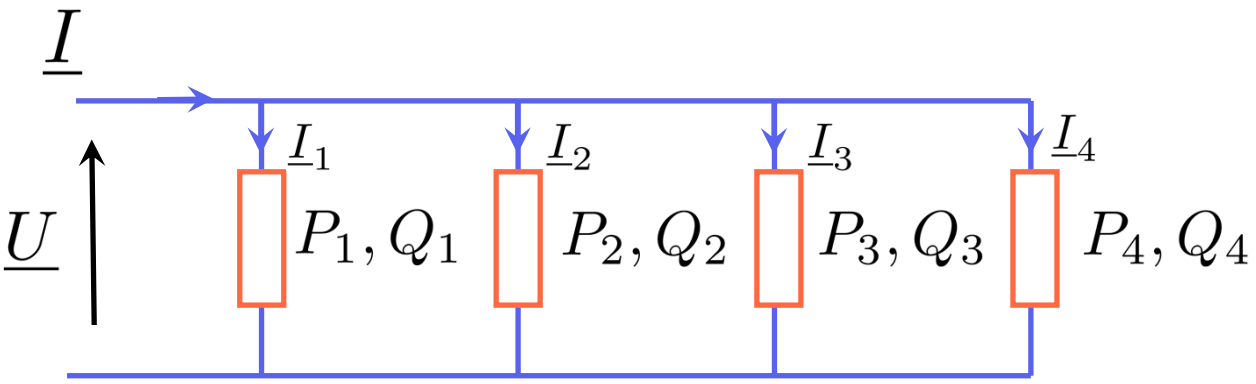
$$\underline{S_T} = P_T + jQ_T$$

$$S_T = \sqrt{P_T^2 + Q_T^2}$$

$$P_T = P_1 + P_2 + P_3 + P_4 = UIcos\varphi$$

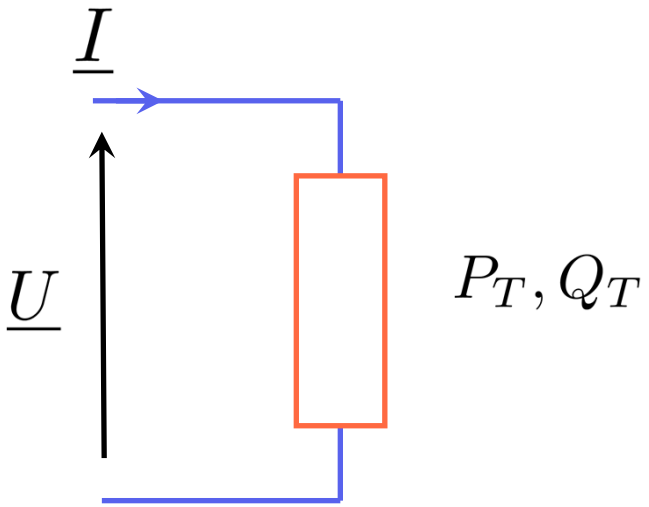
$$Q_T = Q_1 + Q_2 + Q_3 + Q_4 = UIsin\varphi$$

Méthode de Boucherot



$$\underline{I} = \underline{I}_1 + \underline{I}_2 + \underline{I}_3 + \underline{I}_4$$

$$\|\underline{I}\| \neq \|\underline{I}_1\| + \|\underline{I}_2\| + \|\underline{I}_3\| + \|\underline{I}_4\|$$



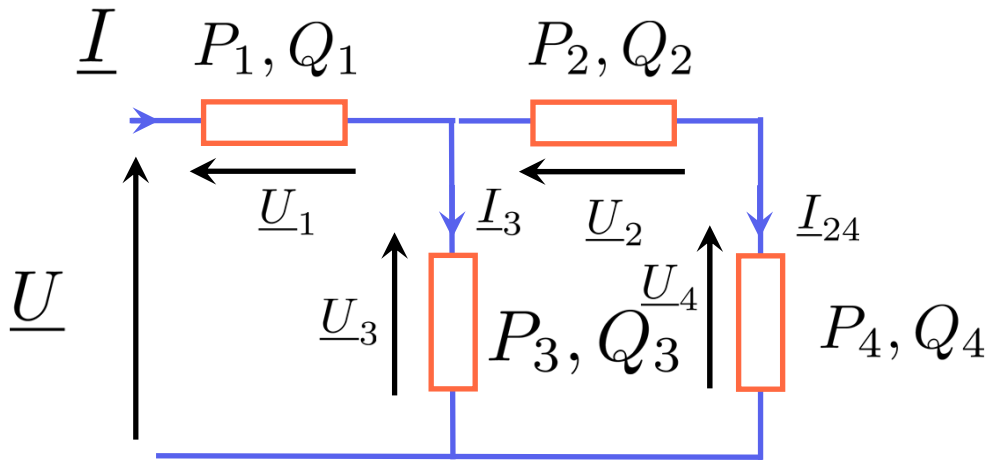
$$\underline{S}_T = P_T + jQ_T$$

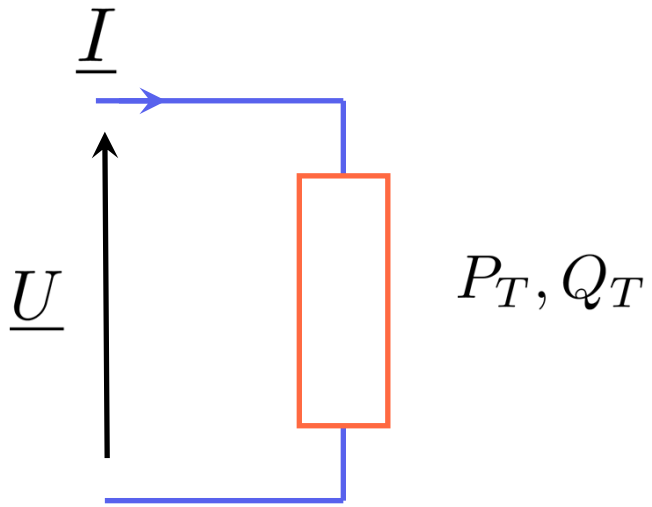
$$S_T = \sqrt{P_T^2 + Q_T^2}$$

$$P_T = P_1 + P_2 + P_3 + P_4 = UIcos\varphi$$

$$Q_T = Q_1 + Q_2 + Q_3 + Q_4 = UIsin\varphi$$

Méthode de Boucherot





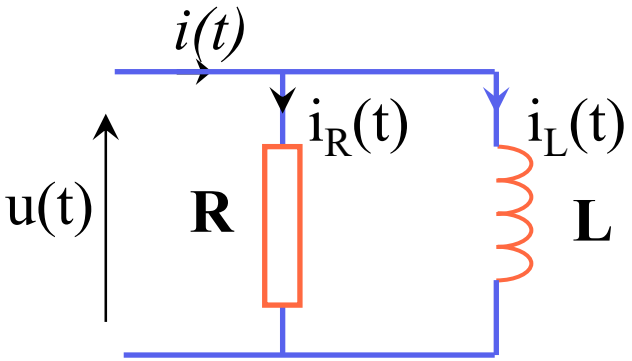
$$\underline{S}_T = P_T + jQ_T$$

$$S_T = \sqrt{P_T^2 + Q_T^2}$$

$$P_T = P_1 + P_2 + P_3 + P_4 = UIcos\varphi$$

$$Q_T = Q_1 + Q_2 + Q_3 + Q_4 = UIsin\varphi$$

Notation complexe : Exemple d'association de dipôle

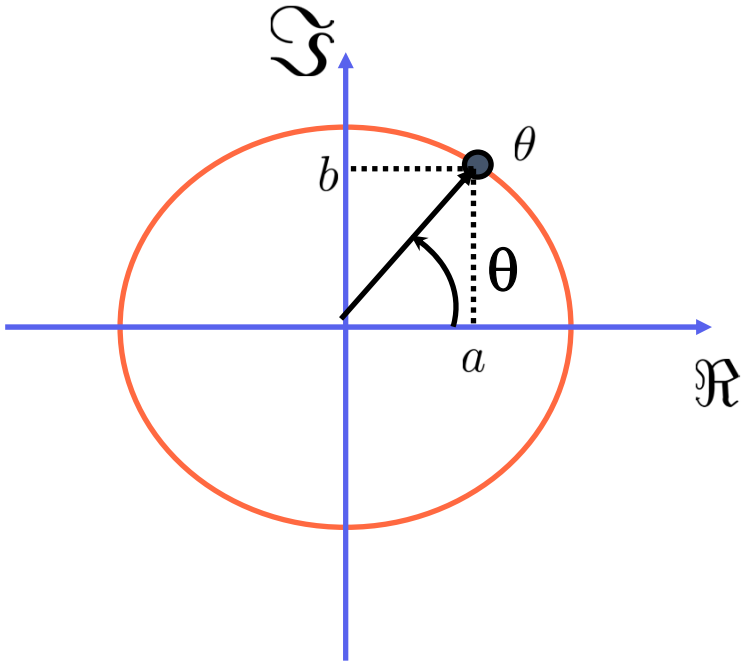


si $u(t) = U_{max}\ sin\omega t$

$$\Rightarrow i_R(t) = \frac{U_{max}}{R}\ sin\omega t \text{ et } i_L(t) = -\frac{U_{max}}{L\omega}\ cos\omega t$$

$$\Rightarrow i(t) = U_{max}(\frac{1}{R}\ sin\omega t - \frac{1}{L\omega}\ cos\omega t)$$

Pour manipuler ces signaux il est nécessaire d'employer la trigonométrie complexe



$$\theta = \omega t$$

$$a + jb = M(cos\theta + jsin\theta) = Me^{j\theta}$$

Avec M le module $M = \sqrt{a^2 + b^2}$

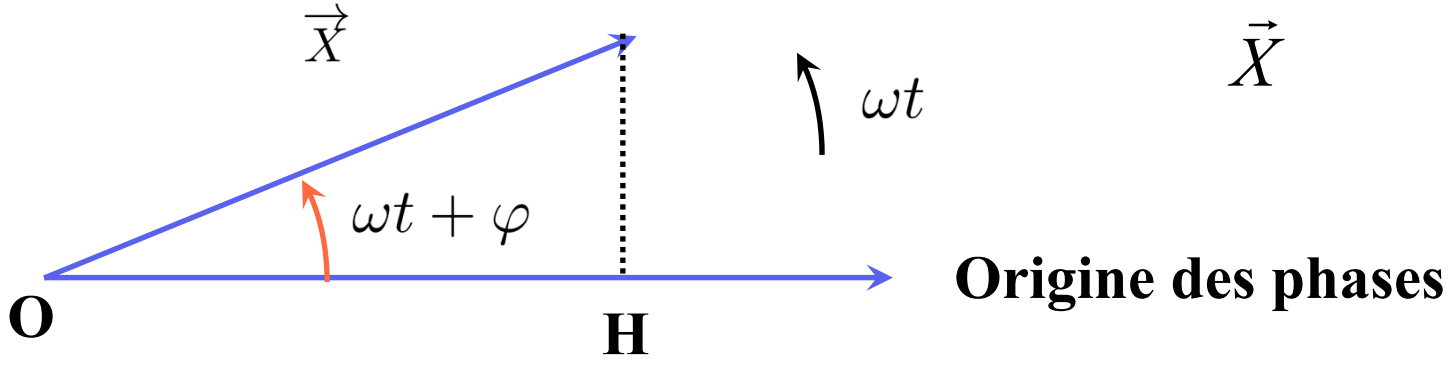
Et l'argument $\theta = arctan\frac{b}{a}$

Représentation vectorielle
Diagramme de Fresnel

$$x(t) = X_M cos(\omega t + \varphi)$$

$$\left\| \vec{X} \right\| = X_M$$

$$\left\| \overrightarrow{OH} \right\| = \left\| \vec{X} \right\| cos(\omega t + \varphi)$$



La projection du vecteur de Fresnel sur l'axe des origines à un instant donné est égale à la valeur instantanée de la fonction associée

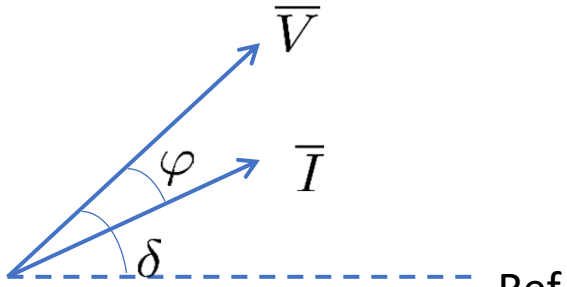
Par convention , on représente les vecteurs de Fresnel à l'instant t=0

Équivalent à la représentation dans le plan complexe

$$x(t) = X_M cos(\omega t + \varphi) = \Re\left[X_M cos(\omega t + \varphi) + j X_M sin(\omega t + \varphi)\right]$$

$$= \Re\left[X_M e^{j(\omega t + \varphi)}\right]$$

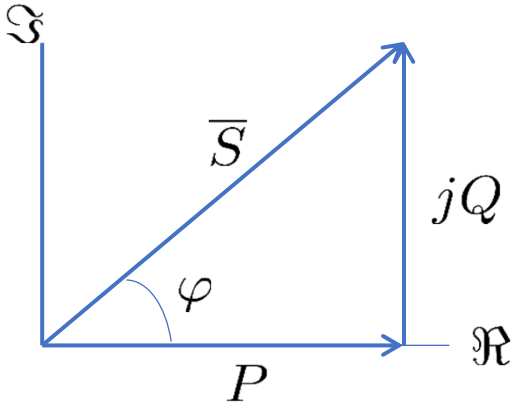
Rappel notation complexe



$$\underline{V} = Ve^{j\delta} = V(cos\delta + jsin\delta) = V\angle\delta$$

$$\underline{I} = Ie^{j(\delta-\varphi)} = I[(cos(\delta-\varphi) + jsin(\delta-\varphi)] = I\angle(\delta-\varphi)$$

Puissances complexes

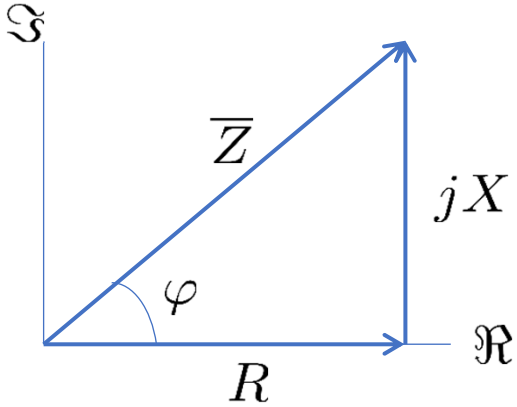


$$\underline{S} = P + jQ = VI(cos\varphi + jsin\varphi) = VIe^{j\varphi} = S\angle\varphi$$

Attention $S = VI$ mais $\underline{S} = \underline{V} \cdot \underline{I}^*$

En effet : $\underline{V} \cdot \underline{I}^* = Ve^{j\delta}Ie^{j(\varphi-\delta)} = VIe^{j\varphi}$

Impédance complexes

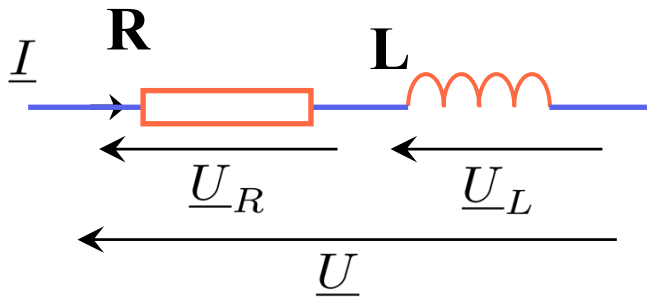


$$\underline{Z} = R + jX = Z(cos\varphi + jsin\varphi) = Ze^{j\varphi} = Z\angle\varphi$$

$$\underline{V} = R \cdot \underline{I} + jX \cdot \overline{I} = (R + jX)\underline{I} = \underline{Z} \cdot \underline{I}$$

Association de dipôle : Circuit RL série

Schéma électrique



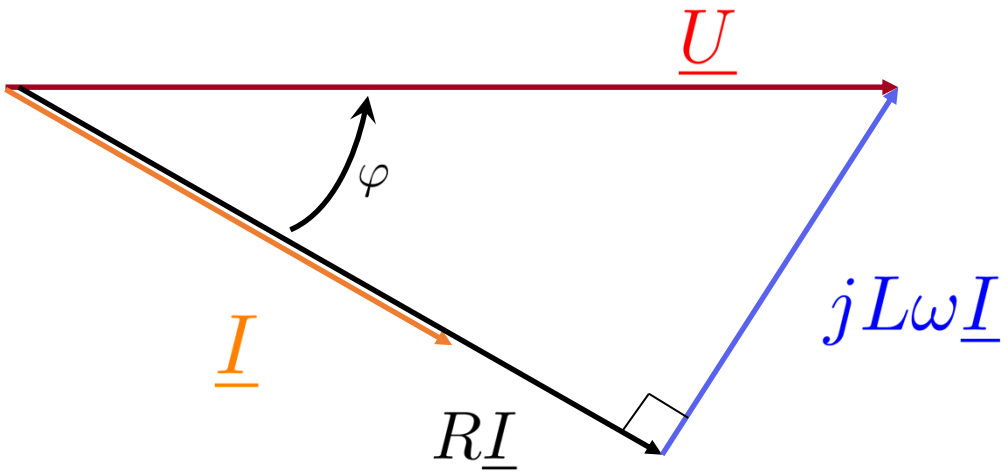


Diagramme vectoriel

Modèle mathématique

$$\underline{U} = \underline{U_R} + \underline{U_L} = R\underline{I} + jL\omega\underline{I}$$

$$\underline{Z} = R + jL\omega$$

$$Z = \sqrt{R^2 + (L\omega)^2}$$

$$\varphi = arctan\frac{L\omega}{R}$$

$$\underline{Z} = Ze^{j\varphi}$$

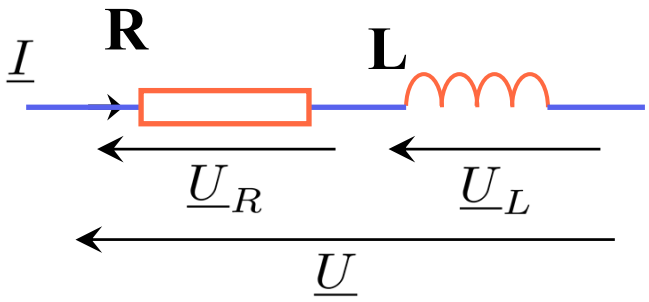
$$\underline{U} = \underline{Z} \cdot \underline{I}$$

$$Ue^{j\omega t} = Ze^{j\varphi} \cdot Ie^{j(\omega t - \varphi)}$$

$$I = \frac{U}{\sqrt{R^2 + (L\omega)^2}}$$

$$\underline{I} = Ie^{j(\omega t - \varphi)}$$

Schéma électrique



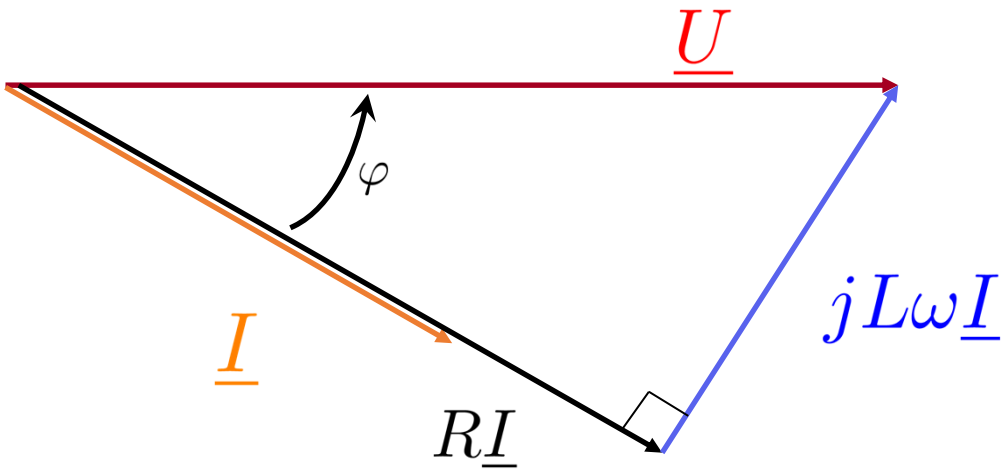


Diagramme vectoriel

Les puissances

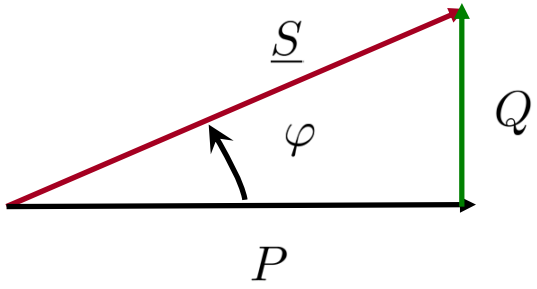
$$P = U_R I = RI^2 = \frac{U_R^2}{R}$$

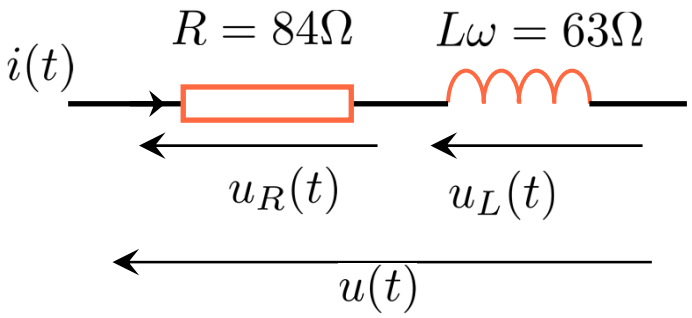
$$Q = U_L I = L\omega I^2 = \frac{U_L^2}{L\omega}$$

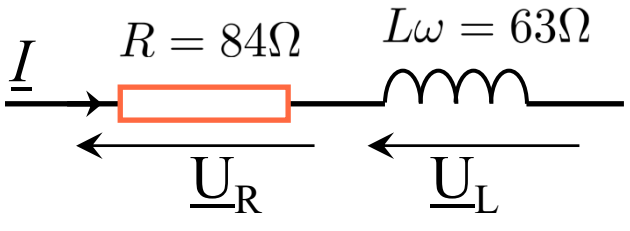
$$\varphi = arctan\frac{Q}{P}$$

$$S = UI = \sqrt{P^2 + Q^2}$$

$$\underline{S} = P + jQ = \underline{U} \cdot \underline{I}^*$$





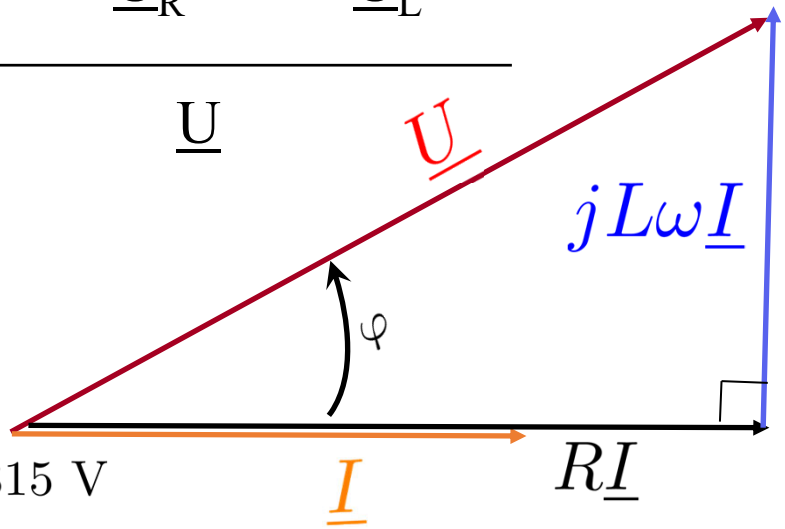


si $i(t) = 5\sqrt{2}\ sin(100\pi t) \Rightarrow \underline{I} = 5$ A

$$u(t) = u_R(t) + u_L(t) = Ri(t) + L\frac{di(t)}{dt}$$

$$u_R(t) = RI_{eff}\sqrt{2}\ sin(100\pi t) \Rightarrow \underline{U}_R = R\underline{I} = 84 \times 5 = 420 \text{ V}$$

$$u_L(t) = L\omega I_{eff}\sqrt{2}\ cos(100\pi t) \Rightarrow \underline{U}_L = jL\omega\underline{I} = j63 \times 5 = j315 \text{ V}$$



$$\underline{U} = \underline{U}_R + \underline{U}_L = (R + jL\omega)\underline{I} = (84 + j63) \times 5 = 420 + j315$$

$$\underline{U} = 525e^{j0.6435} = 525\angle 36.86^o$$

$$P = U_R I = RI^2 = \frac{U_R^2}{R}$$

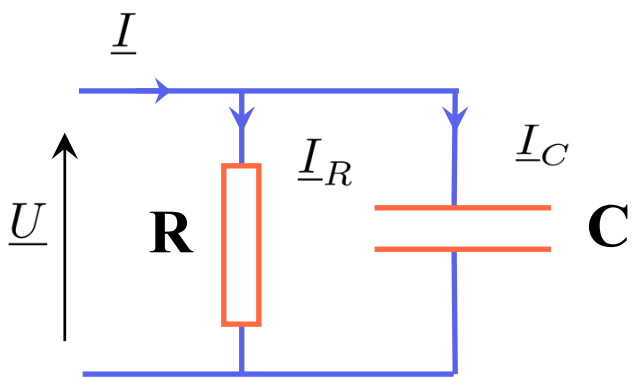
$$Q = U_L I = L\omega I^2 = \frac{U_L^2}{L\omega}$$

$$S = UI = \sqrt{P^2 + Q^2}$$

$$S = 525 \times 5 = \sqrt{2100^2 + 1575^2} = 2625 \text{ VA}$$

Association de dipôle : Circuit RC parallèle

Schéma électrique



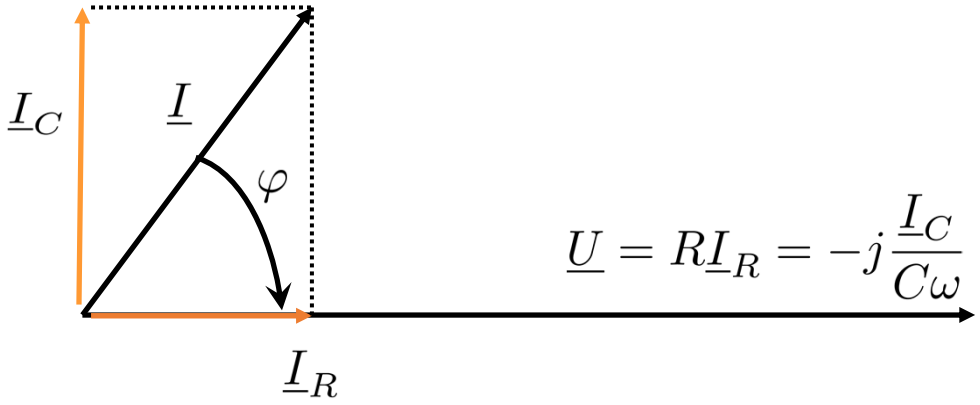


Diagramme vectoriel

Modèle mathématique

$$\underline{I} = \underline{I}_R + \underline{I}_C = \frac{\underline{U}}{R} + jC\omega\underline{U}$$

$$\underline{Y} = \frac{1}{R} + jC\omega$$

$$Y = \sqrt{\frac{1}{R^2} + (C\omega)^2}$$

$$\varphi = arctanRC\omega$$

$$\underline{Y} = Ye^{j\varphi}$$

$$\underline{I} = \underline{Y} \cdot \underline{U}$$

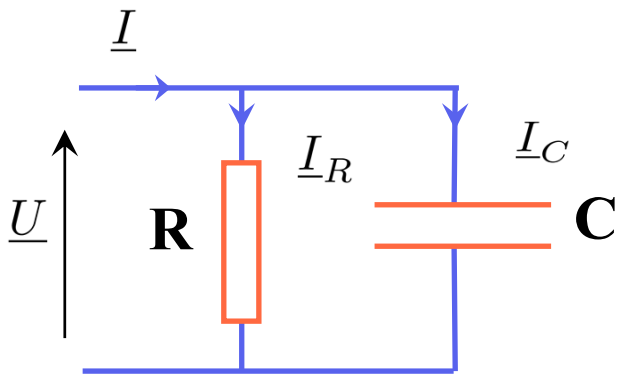
$$Ie^{(j\omega t+\varphi)} = Ye^{j\varphi} \cdot Ue^{j\omega t}$$

$$I = \sqrt{\frac{1}{R^2} + (C\omega)^2} \cdot U$$

$$\underline{I} = Ie^{j(\omega t+\varphi)}$$

Association de dipôle : Circuit RC parallèle

Schéma électrique



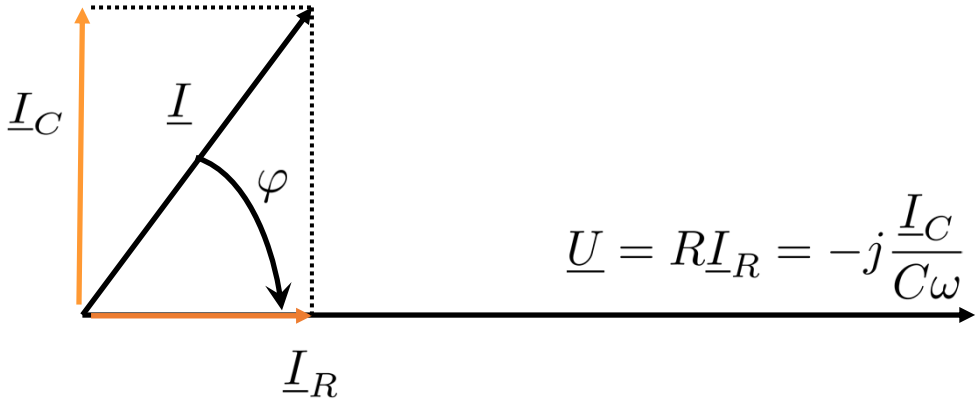


Diagramme vectoriel

Les puissances

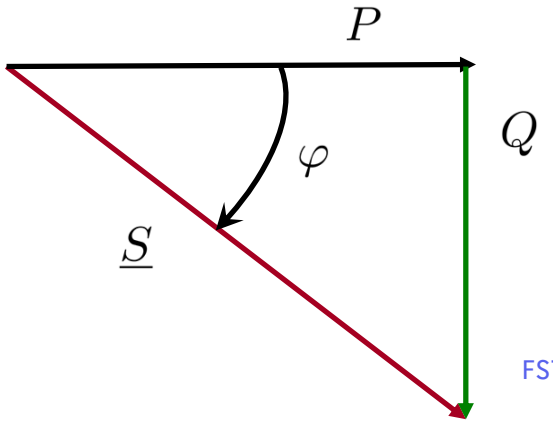
$$P = U \cdot I_R = R \cdot I_R^2 = \frac{U^2}{R}$$

$$Q = -U \cdot I_C = -C\omega U^2 = -\frac{I_C^2}{L\omega}$$

$$\varphi = -arctan\frac{Q}{P}$$

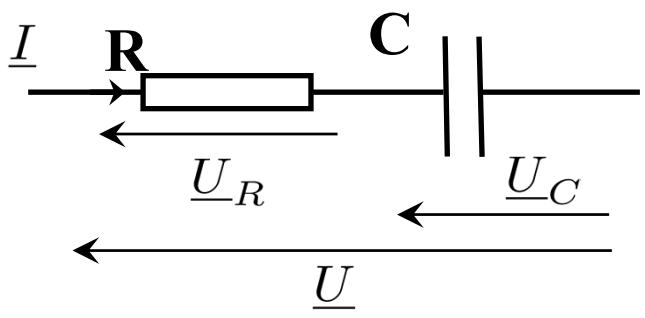
$$S = UI = \sqrt{P^2 + Q^2}$$

$$\underline{S} = P + jQ = \underline{U} \cdot \underline{I}^*$$



Association de dipôle : Circuit RC série

Schéma électique



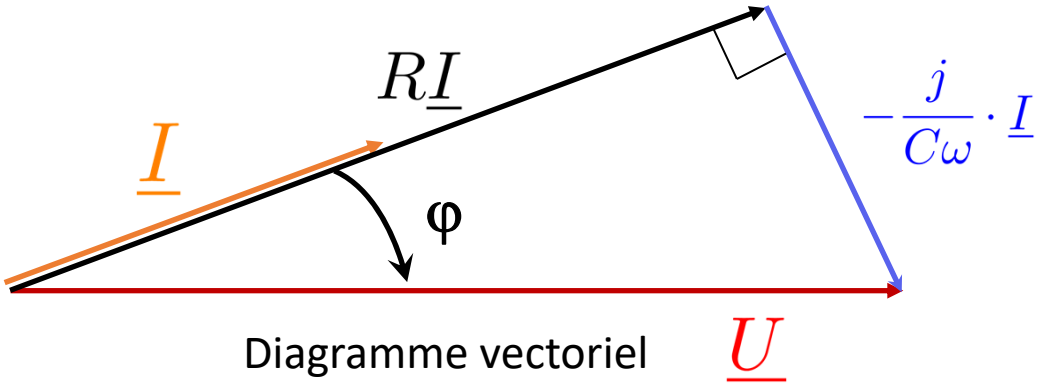


Diagramme vectoriel

Modèle mathématique

$$\underline{U} = \underline{U}_R + \underline{U}_C = R\underline{I} + \frac{\underline{I}}{jC\omega}$$

$$\underline{Z} = R - \frac{j}{C\omega}$$

$$Z = \sqrt{R^2 + (\frac{1}{C\omega})^2}$$

$$\varphi = -arctan\frac{1}{RC\omega}$$

$$\underline{Z} = Ze^{-j\varphi}$$

$$\underline{U} = \underline{Z}\cdot\underline{I}$$

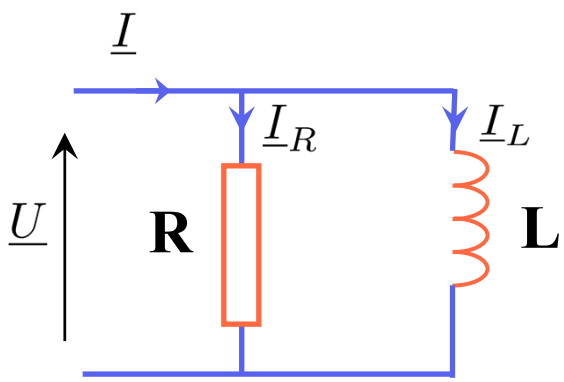
$$Ue^{j\omega t} = Ze^{-j\varphi}\cdot Ie^{j(\omega t+\varphi)}$$

$$I = \frac{U}{\sqrt{R^2 + (\frac{1}{C\omega})^2}}$$

$$\underline{I} = Ie^{j(\omega t+\varphi)}$$

Association de dipôle : Circuit RL parallèle

Schéma électrique



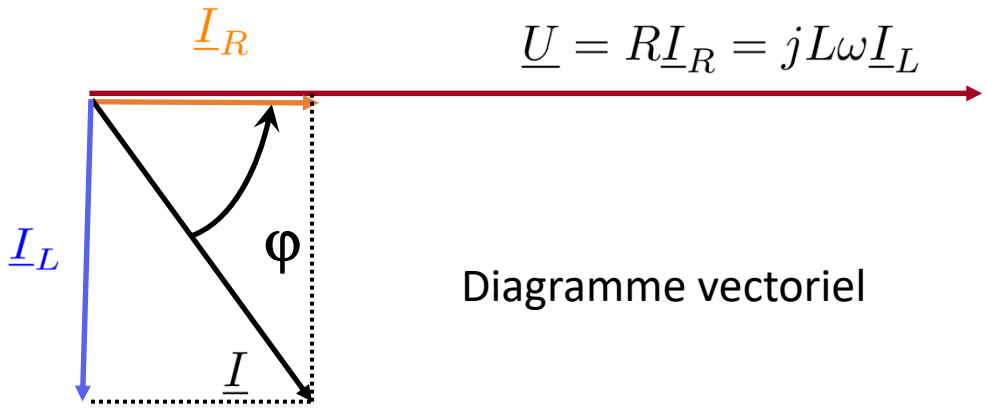


Diagramme vectoriel

Modèle mathématique

$$\underline{I} = \underline{I}_R + \underline{I}_L = \frac{\underline{U}}{R} + \frac{\underline{U}}{jL\omega}$$

$$\underline{Y} = \frac{1}{R} + \frac{1}{jL\omega}$$

$$Y = \sqrt{\frac{1}{R^2} + \frac{1}{(L\omega)^2}}$$

$$\varphi = -arctan\frac{R}{L\omega}$$

$$\underline{Y} = Ye^{-j\varphi}$$

$$\underline{I} = \underline{Y} \cdot \underline{U}$$

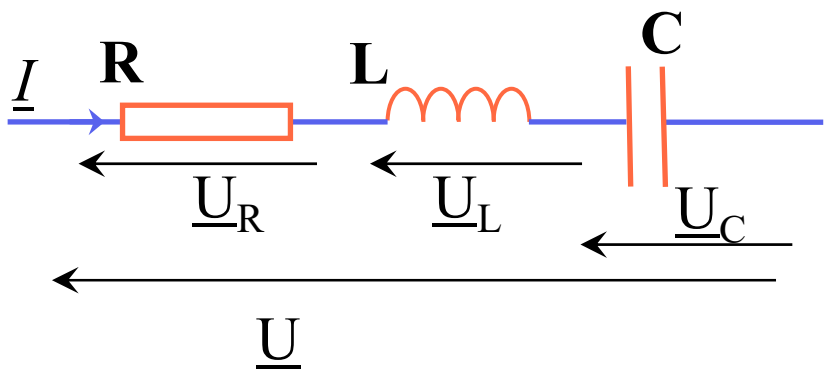
$$Ie^{(j\omega t-\varphi)} = Ye^{-j\varphi} \cdot Ue^{j\omega t}$$

$$I = \sqrt{\frac{1}{R^2} + \frac{1}{(L\omega)^2}} \cdot U$$

$$\underline{I} = Ie^{j(\omega t+\varphi)}$$

Association de dipôle : Circuit RLC série

Schéma électrique



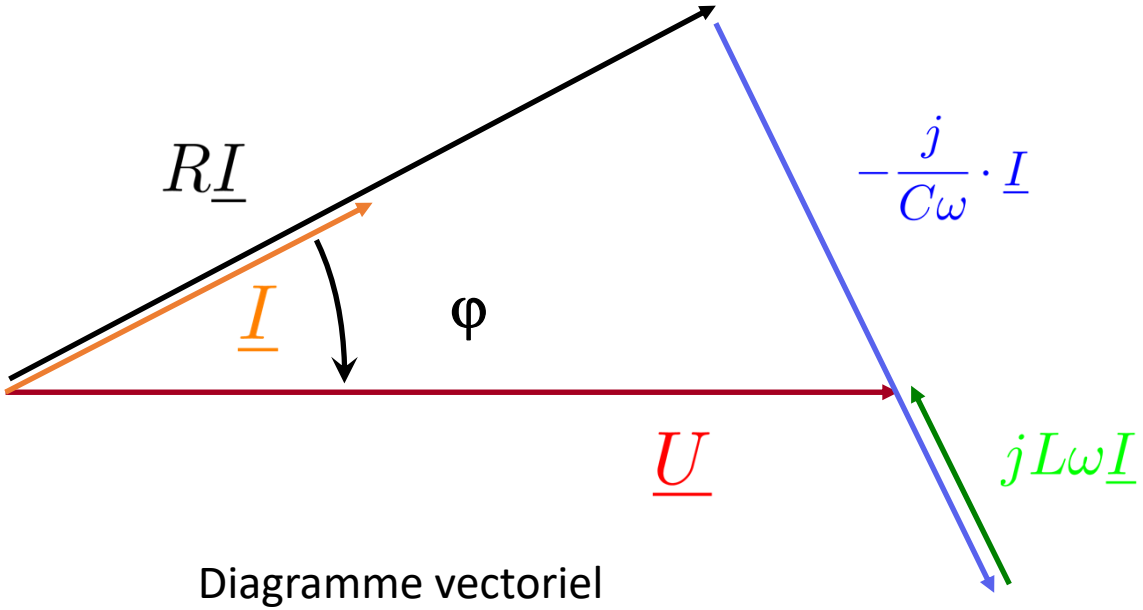


Diagramme vectoriel

$$L\omega < \frac{1}{C\omega}$$

Modèle mathématique

$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = R\underline{I} + jL\omega\underline{I} + \frac{\underline{I}}{jC\omega}$$

$$\underline{Z} = R + j(L\omega - \frac{1}{C\omega})$$

$$Z = \sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$$

$$\varphi = arctan\frac{L\omega - \frac{1}{C\omega}}{R}$$

$$\underline{Z} = Ze^{j\varphi}$$

$$\underline{U} = \underline{Z}\cdot\underline{I}$$

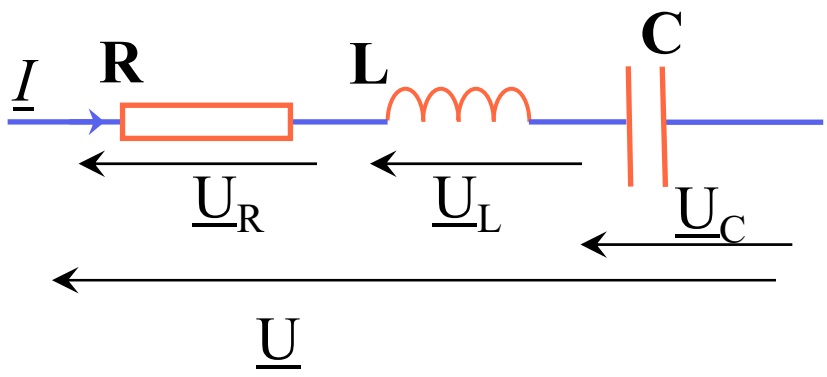
$$Ue^{j\omega t} = Ze^{j\varphi}\cdot Ie^{j(\omega t-\varphi)}$$

$$I = \frac{U}{\sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}}$$

$$\underline{I} = Ie^{j(\omega t-\varphi)}$$

Association de dipôle : Circuit RLC série

Schéma électrique



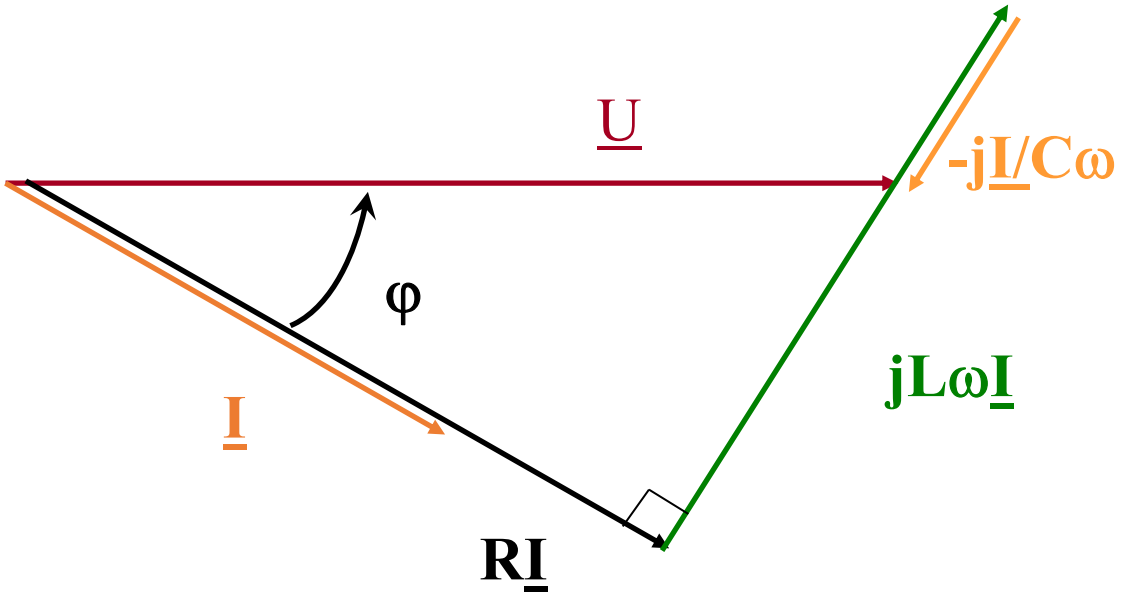


Diagramme vectoriel

$$L\omega > \frac{1}{C\omega}$$

Modèle mathématique

$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = R\underline{I} + jL\omega\underline{I} + \frac{\underline{I}}{jC\omega}$$

$$\underline{Z} = R + j(L\omega - \frac{1}{C\omega})$$

$$Z = \sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$$

$$\varphi = arctan\frac{L\omega - \frac{1}{C\omega}}{R}$$

$$\underline{Z} = Ze^{j\varphi}$$

$$\underline{U} = \underline{Z} \cdot \underline{I}$$

$$Ue^{j\omega t} = Ze^{j\varphi} \cdot Ie^{j(\omega t - \varphi)}$$

$$I = \frac{U}{\sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}}$$

$$\underline{I} = Ie^{j(\omega t - \varphi)}$$

Schéma électrique

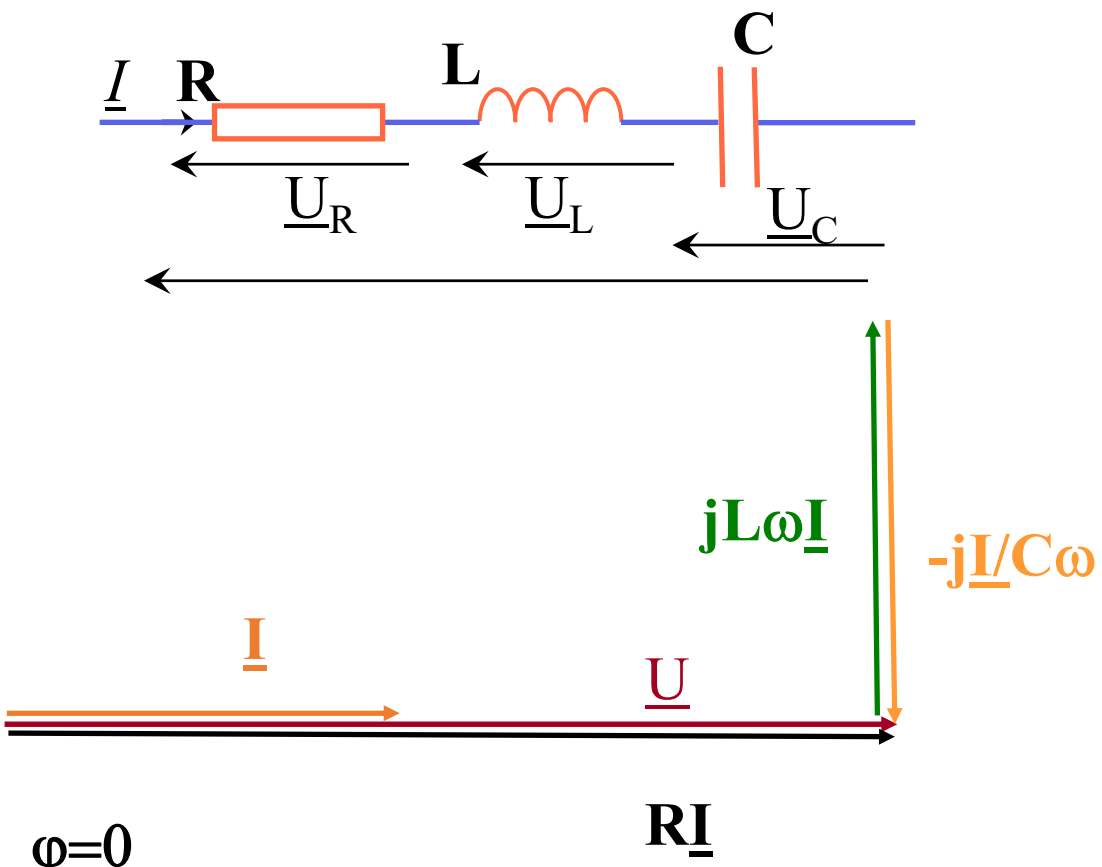


Diagramme vectoriel

$$L\omega = \frac{1}{C\omega} \qquad Q_L + Qc = 0$$

Modèle mathématique

$$\underline{U} = \underline{U}_R + \underline{U}_L + \underline{U}_C = R\underline{I} + jL\omega\underline{I} + \frac{I}{jC\omega}$$

$$\underline{Z} = R + j(L\omega - \frac{1}{C\omega})$$

$$Z = \sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}$$

$$\varphi = arctan\frac{L\omega - \frac{1}{C\omega}}{R} = 0$$

$$\underline{Z} = Ze^{j0}$$

$$\underline{U} = \underline{Z} \cdot \underline{I}$$

$$Ue^{j\omega t} = Ze^{j0} \cdot Ie^{j(\omega t)}$$

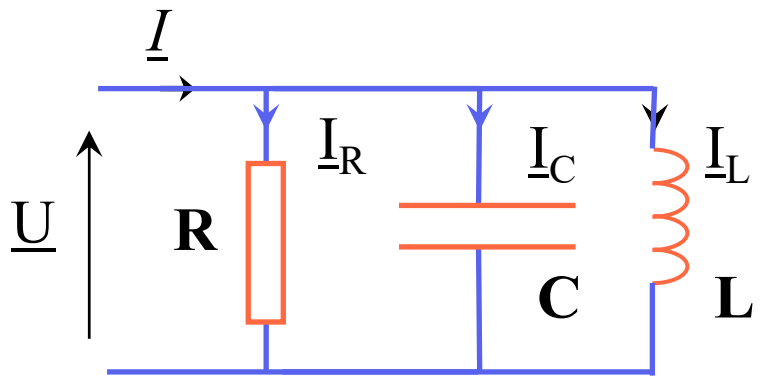
$$I = \frac{U}{\sqrt{R^2 + (L\omega - \frac{1}{C\omega})^2}} = \frac{U}{R}$$

$$\underline{I} = Ie^{j(\omega t)}$$

Association de dipôle : Circuit RLC Parallèle résonant

Université de Lille

Schéma électrique



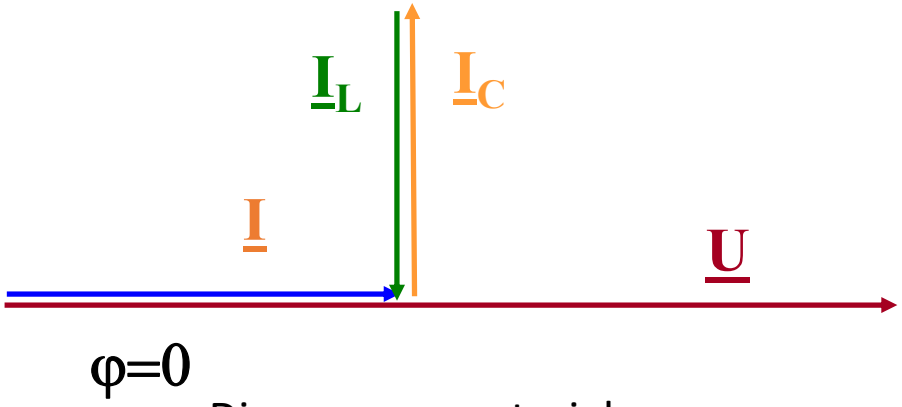


Diagramme vectoriel

$$Q_L + Qc = 0 \qquad L\omega = \frac{1}{C\omega}$$

Modèle mathématique

$$\underline{I} = \underline{I}_R + \underline{I}_L + \underline{I}_C = \frac{\underline{U}}{R} + \frac{\underline{U}}{jL\omega} + jC\omega\underline{U}$$

$$\underline{Y} = \frac{1}{R} + \frac{1}{jL\omega} + jC\omega\underline{U}$$

$$Y = \sqrt{\frac{1}{R^2} + (C\omega - \frac{1}{L\omega})^2}$$

$$\varphi = arctanR(C\omega - \frac{1}{L\omega})$$

$$\underline{Y} = Ye^{j\varphi}$$

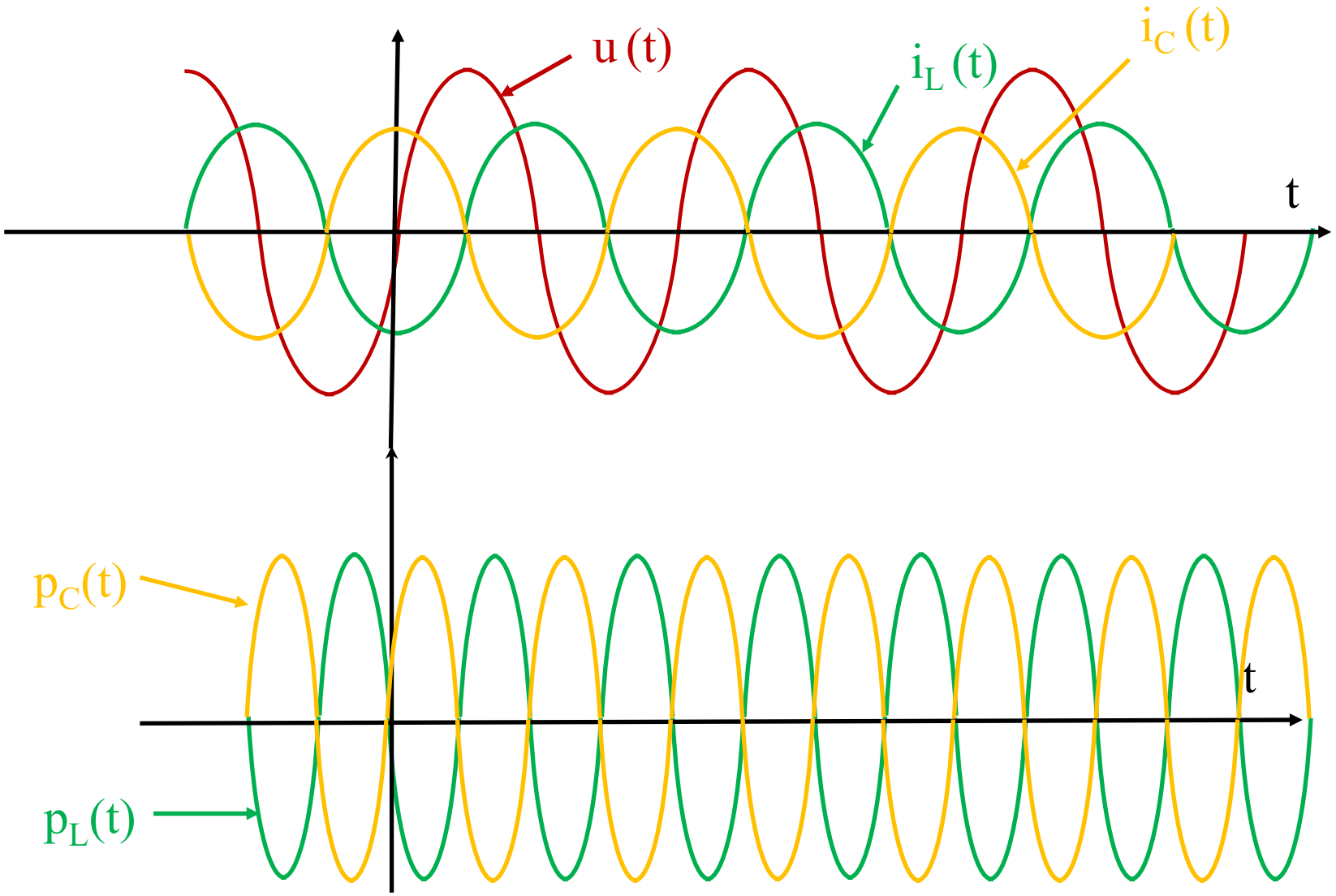
$$\underline{I} = \underline{Y} \cdot \underline{U}$$

$$Ie^{(j\omega t+\varphi)} = Ye^{j\varphi} \cdot Ue^{j\omega t}$$

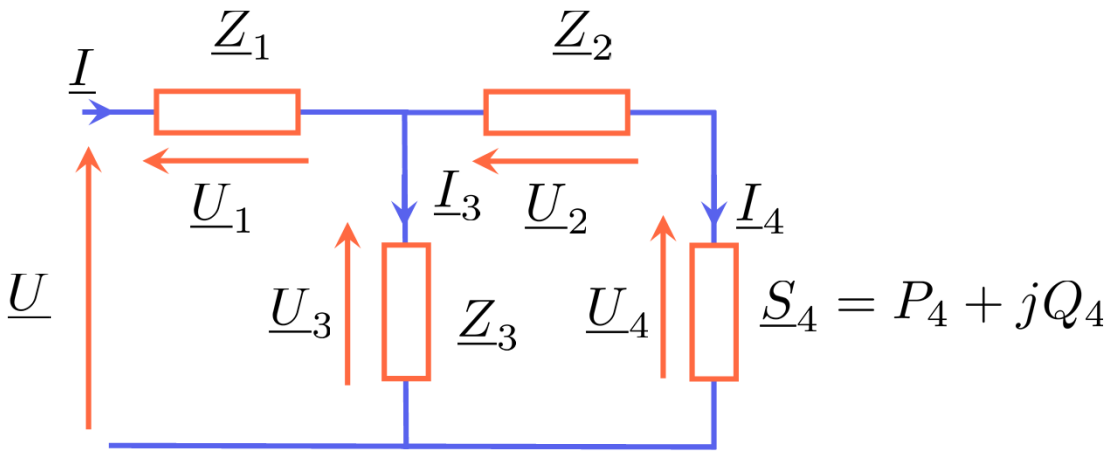
$$I = \sqrt{\frac{1}{R^2} + (C\omega - \frac{1}{L\omega})^2} \cdot U$$

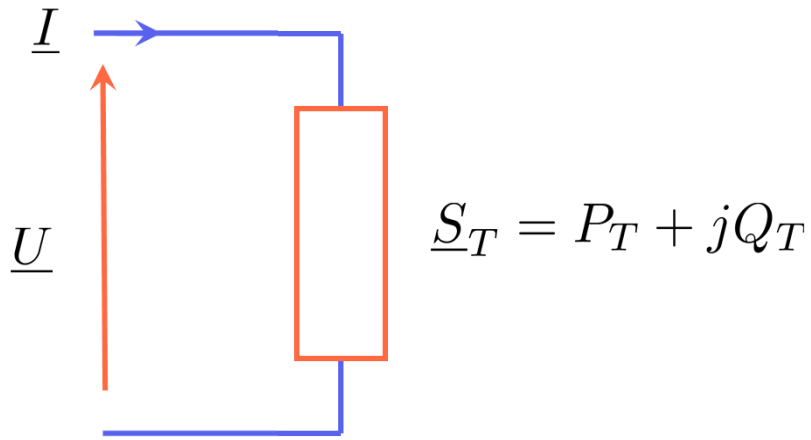
$$\underline{I} = Ie^{j(\omega t+\varphi)}$$

Association de dipôle : Echange d'énergie circuit résonnant



Applications de la Méthode de Boucherot





$$\underline{Z}_1 = \underline{Z}_2 = 1 + 2j \quad \underline{U}_4 = 230$$

$$\underline{S}_4 = P_4 + jQ_4 = 1000 + j500$$

$$\underline{Z}_3 = R_3 // C_3$$

$$R_3 = 40\ \Omega$$

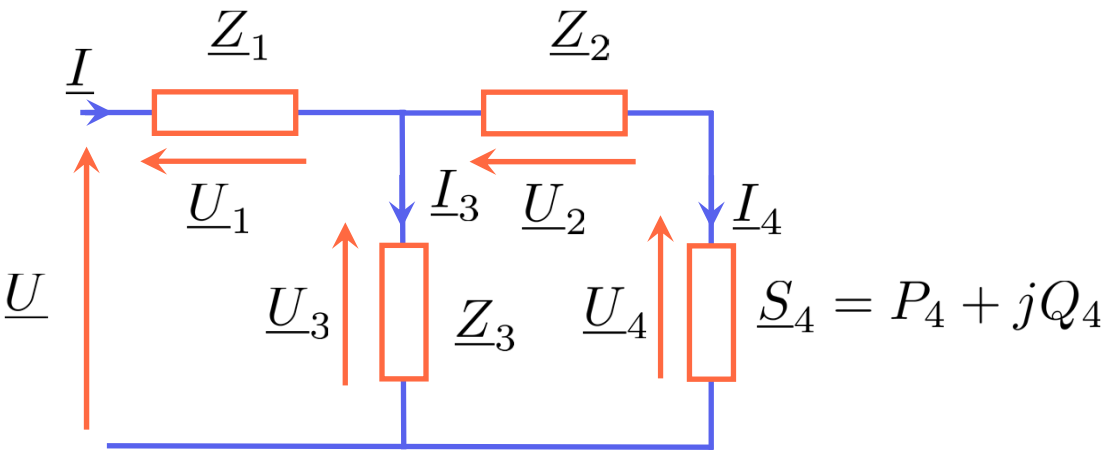
$$\frac{1}{C_3\omega} = 50\ \Omega$$

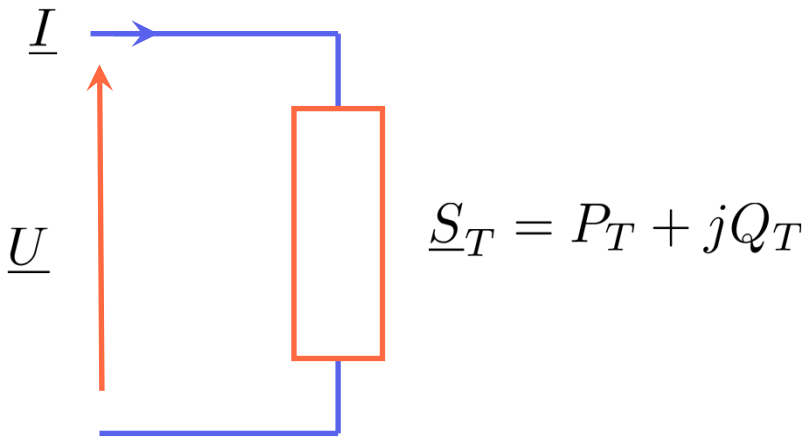
$$\underline{Z}_4 = R_4 // L_4$$

$$R_4 = \frac{U_4^2}{P_4} = \frac{230^2}{1000} = 52.9\ \Omega$$

$$L_4\omega = \frac{U_4^2}{Q_4} = \frac{230^2}{500} = 105.8\ \Omega$$

Applications de la Méthode de Boucherot





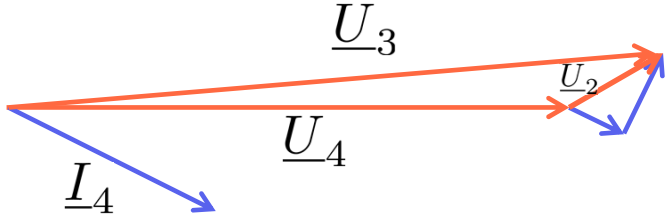
$$\underline{I}_4^* = \frac{\underline{S}_4}{\underline{U}_4} \Rightarrow \underline{I}_4 = \frac{\underline{S}_4^*}{\underline{U}_4^*} = \frac{1000 - j500}{230} = 4.861\angle -26.56^o$$

$$P_2 = R_2 \times I_4^2 = 1 \times 4.861^2 = 23.62 \text{ W}$$

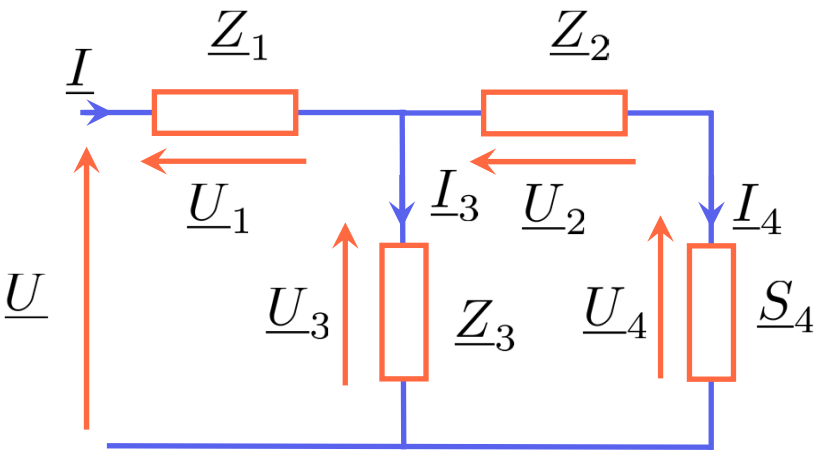
$$Q_2 = L_2\omega \times I_4^2 = 2 \times 4.861^2 = 47.26 \text{ VAr}$$

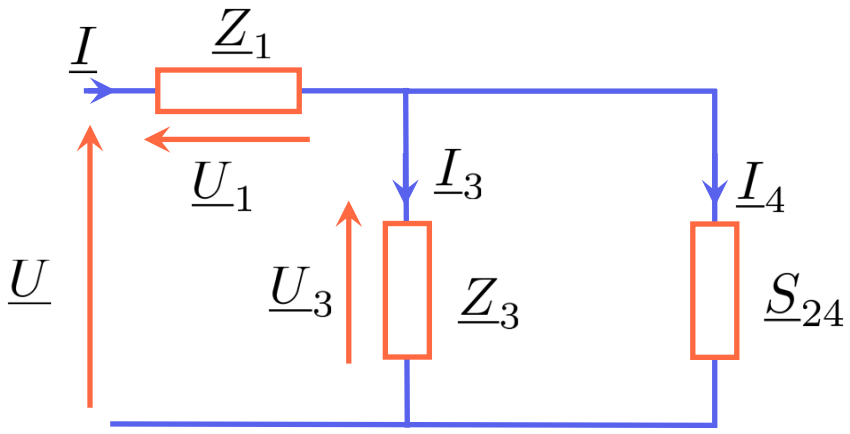
$$\underline{S}_2 = P_2 + jQ_2 = 23.63 + j47.26 = 52.83\angle 63.44^o$$

$$\underline{U}_2 = \frac{\underline{S}_2}{\underline{I}_4^*} = \frac{52.83\angle 63.44^o}{4.861\angle 26.56^o} = 10.868\angle 36.88^o$$



Applications de la Méthode de Boucherot





$$\underline{I}_4 = 4.861\angle -26.56^o$$

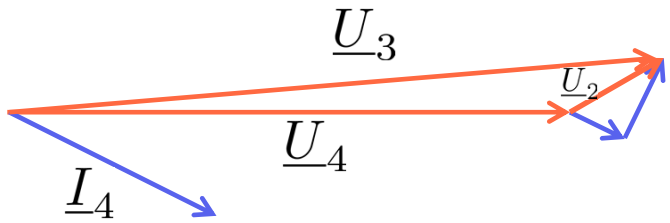
$$P_{24} = P_2 + P_4 = 1023.32 \text{ W}$$

$$Q_{24} = Q_2 + Q_4 = 547.25 \text{ VAr}$$

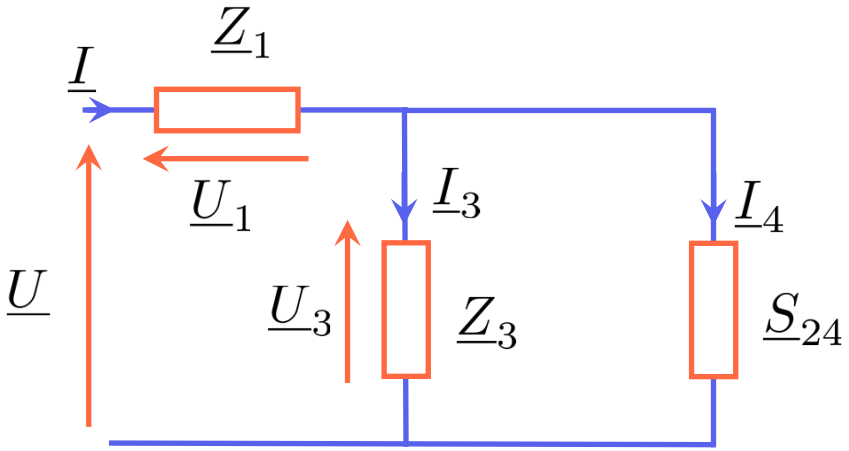
$$\underline{S}_{24} = 1160\angle 28.13^o$$

$$\underline{U}_3 = \frac{\underline{S}_{24}}{\underline{I}_4^*} = \frac{1160\angle 28.13^o}{4.861\angle 26.56^o} = 238.53\angle 1.57^o$$

$$S_3 = U_3 I_3$$



Applications de la Méthode de Boucherot



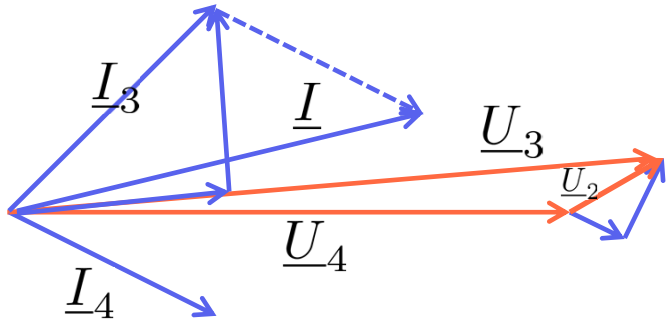
$$\underline{I}_4 = 4.861\angle -26.56^o$$

$$\underline{S}_{24} = 1160\angle 28.13^o$$

$$\underline{U}_3 = 238.53\angle 1.57^o$$

$$\underline{Z}_3 = R_3//C_3 \qquad R_3 = 40\ \Omega$$

$$\frac{1}{C_3\omega} = 50\ \Omega$$



$$P_3 = \frac{U_3^2}{R_3} = \frac{238.53^2}{40} = 1422\ \text{W}$$

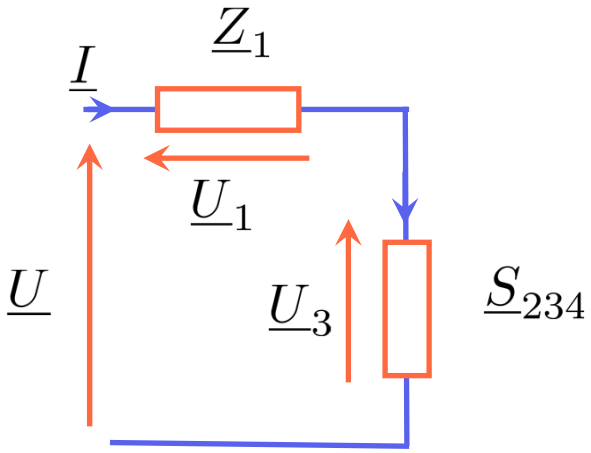
$$Q_3 = -U_3^2 \times C_3\omega = -\frac{238.53^2}{50} = -1137\ \text{VAr}$$

$$S_3 = \sqrt{P_3^2 + Q_3^2} = \sqrt{1422^2 + (-1137)^2} \Rightarrow \underline{S}_3 = 1820\angle -38.64^o$$

$$\underline{I} = \underline{I}_4 + \underline{I}_3$$

$$\underline{I}_3 = \frac{\underline{S}_3^*}{\underline{U}_3^*} = \frac{1820\angle 38.64^o}{238\angle -1.57^o} = 7.68\angle 40.21^o$$

Applications de la Méthode de Boucherot



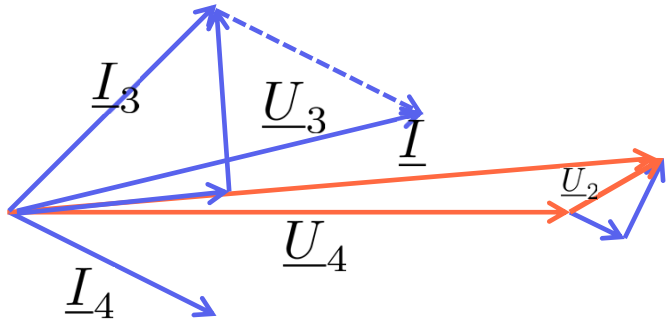
$$\underline{I}_4 = 4.861\angle -26.56^o$$

$$\underline{I}_3 = 7.68\angle 37.07^o$$

$$S_{234} = \sqrt{(P_2 + P_3 + P_4)^2 + (Q_2 + Q_3 + Q_4)^2}$$

$$S_{234} = \sqrt{(1422 + 23 + 1000)^2 + (-1137 + 47 + 500)^2}$$

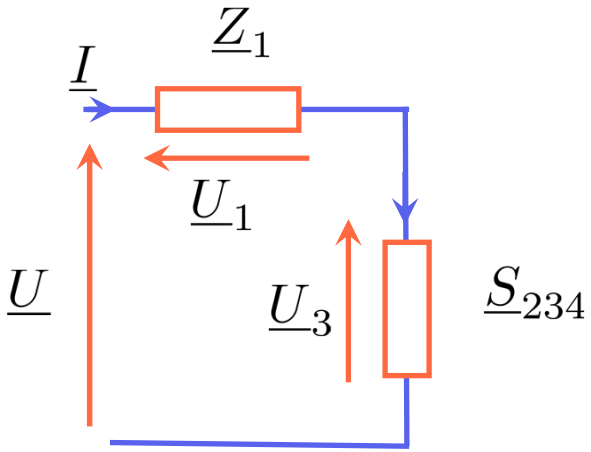
$$\underline{S}_{234} = 2515\angle -13.56^o$$

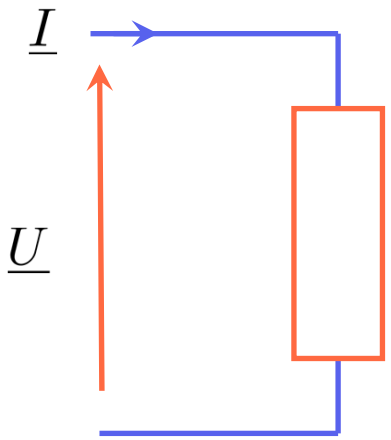


$$\underline{I} = \frac{\underline{S}_{234}^*}{\underline{U}_3^*} = \frac{2515\angle 13.56^o}{238\angle -1.57^o} = 10.53\angle 15.13^o$$

$$\underline{S}_1 = P_1 + jQ_1 = R_1 I^2 + jL_1\omega I^2 = 110.88 + j221.76 = 248\angle 63.45^o$$

Applications de la Méthode de Boucherot





$$\underline{S}_T = P_T + jQ_T$$

$$S_{1234} = \sqrt{(P_1 + P_{234})^2 + (Q_1 + Q_{234})^2}$$

$$S_{1234} = \sqrt{(110 + 2445)^2 + (221 - 590)^2}$$

$$\underline{S}_T = \underline{S}_{1234} = 2582\angle -8.43^o$$

$$\underline{U}_1 = \frac{\underline{S}_1}{\underline{I}^*} = \frac{248\angle 63.44^o}{10.53\angle -15.13^o} = 23.45\angle 78.24^o$$

$$\underline{U} = \frac{\underline{S}_T}{\underline{I}^*} = \frac{248\angle -8.43^o}{10.53\angle -15.13^o} = 245.2\angle 7.1^o$$

